

BFS Traversal

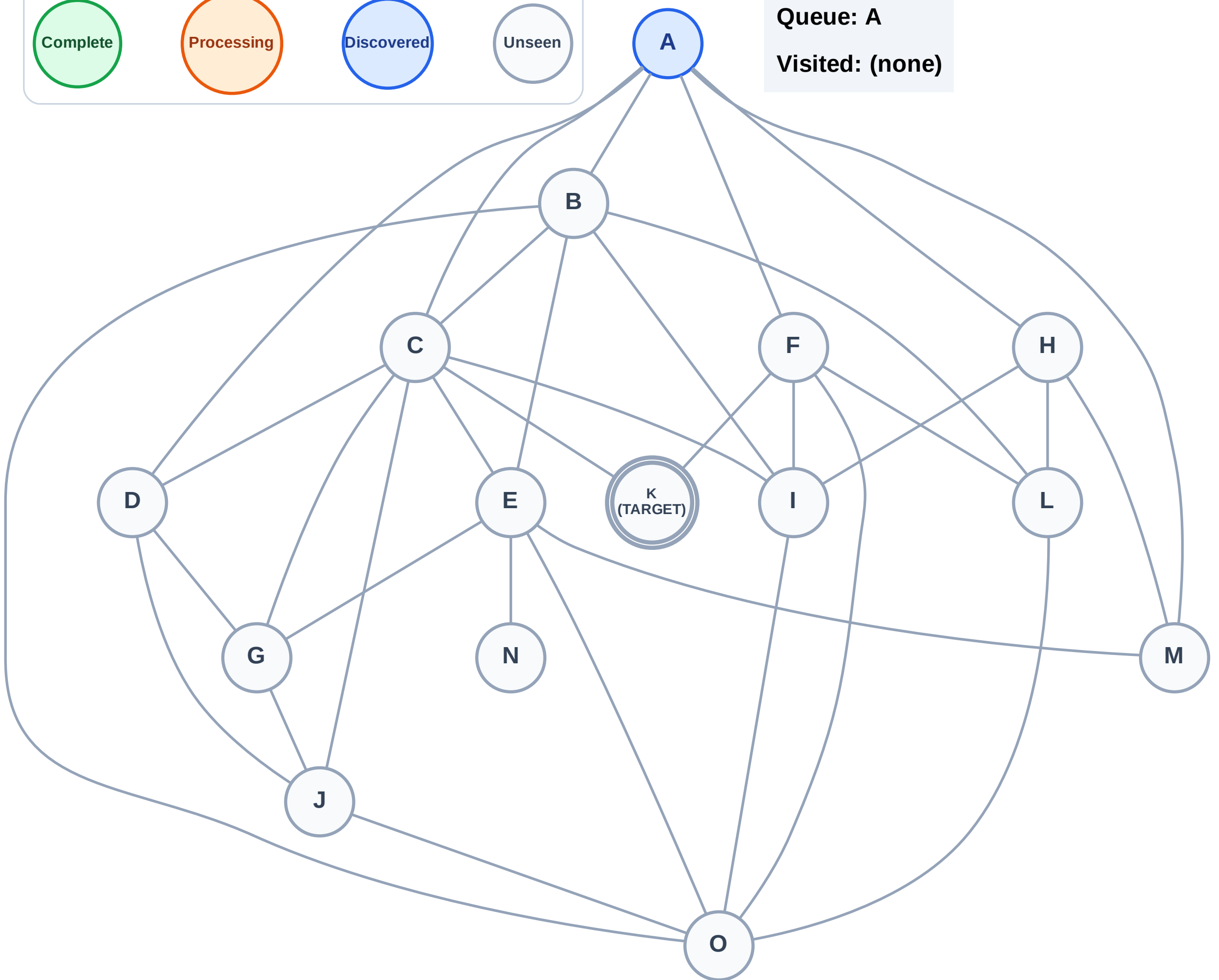
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

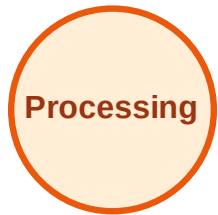
Visited: (none)



BFS Traversal

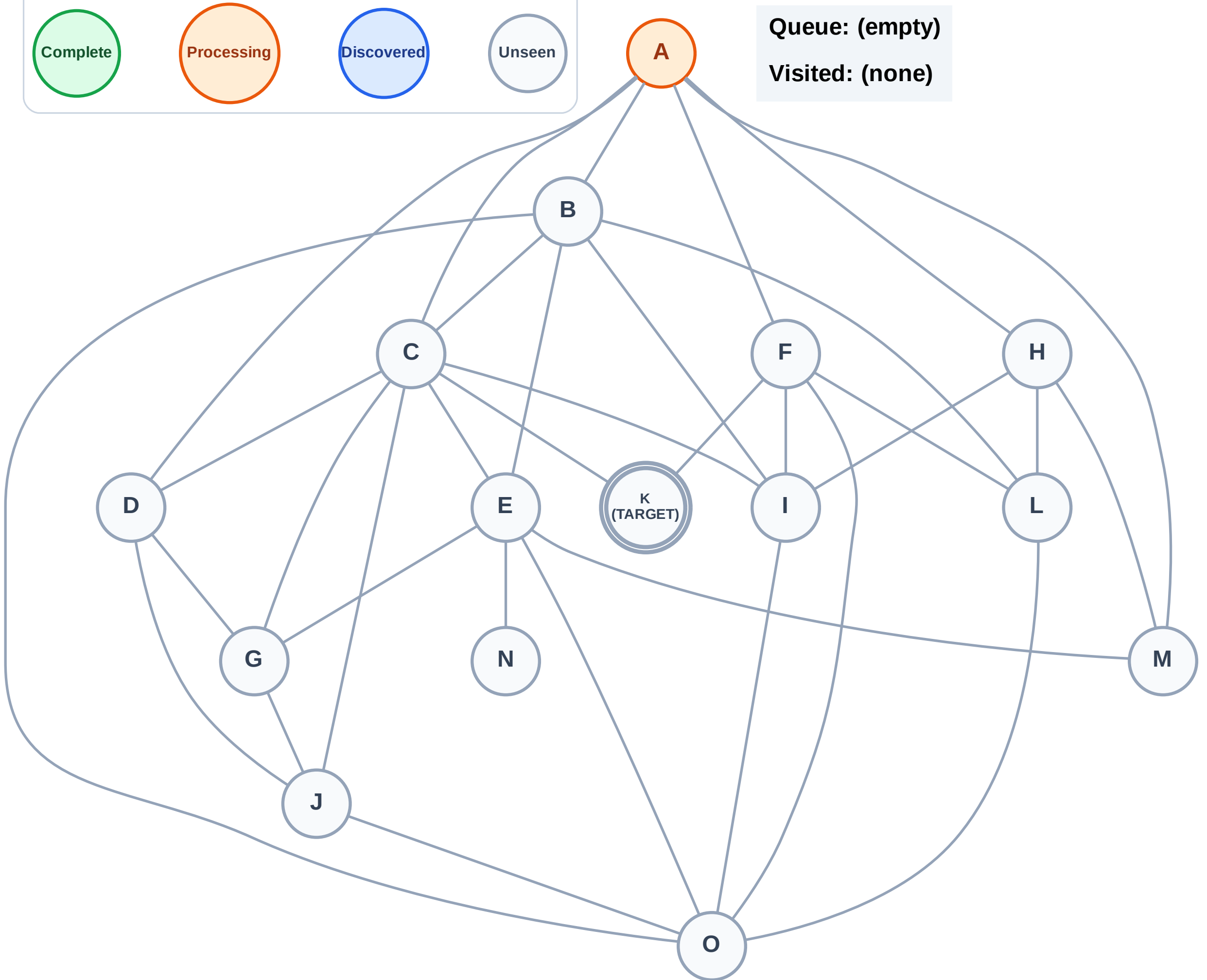
Step 2: Process node A

Legend



Queue: (empty)

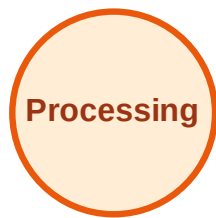
Visited: (none)



BFS Traversal

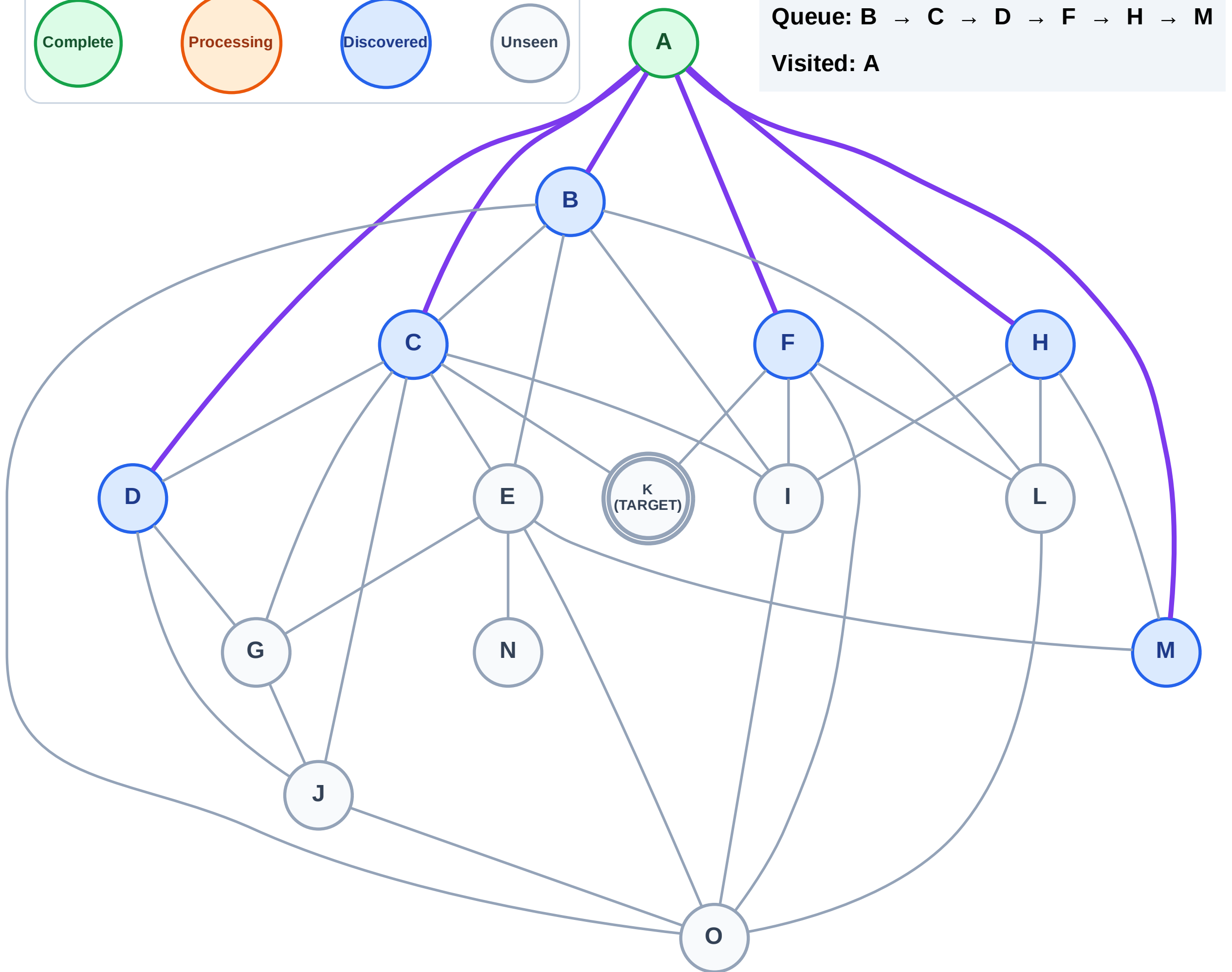
Step 3: Discover B, C, D, F, H, M from A

Legend



Queue: B → C → D → F → H → M

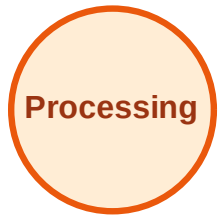
Visited: A



BFS Traversal

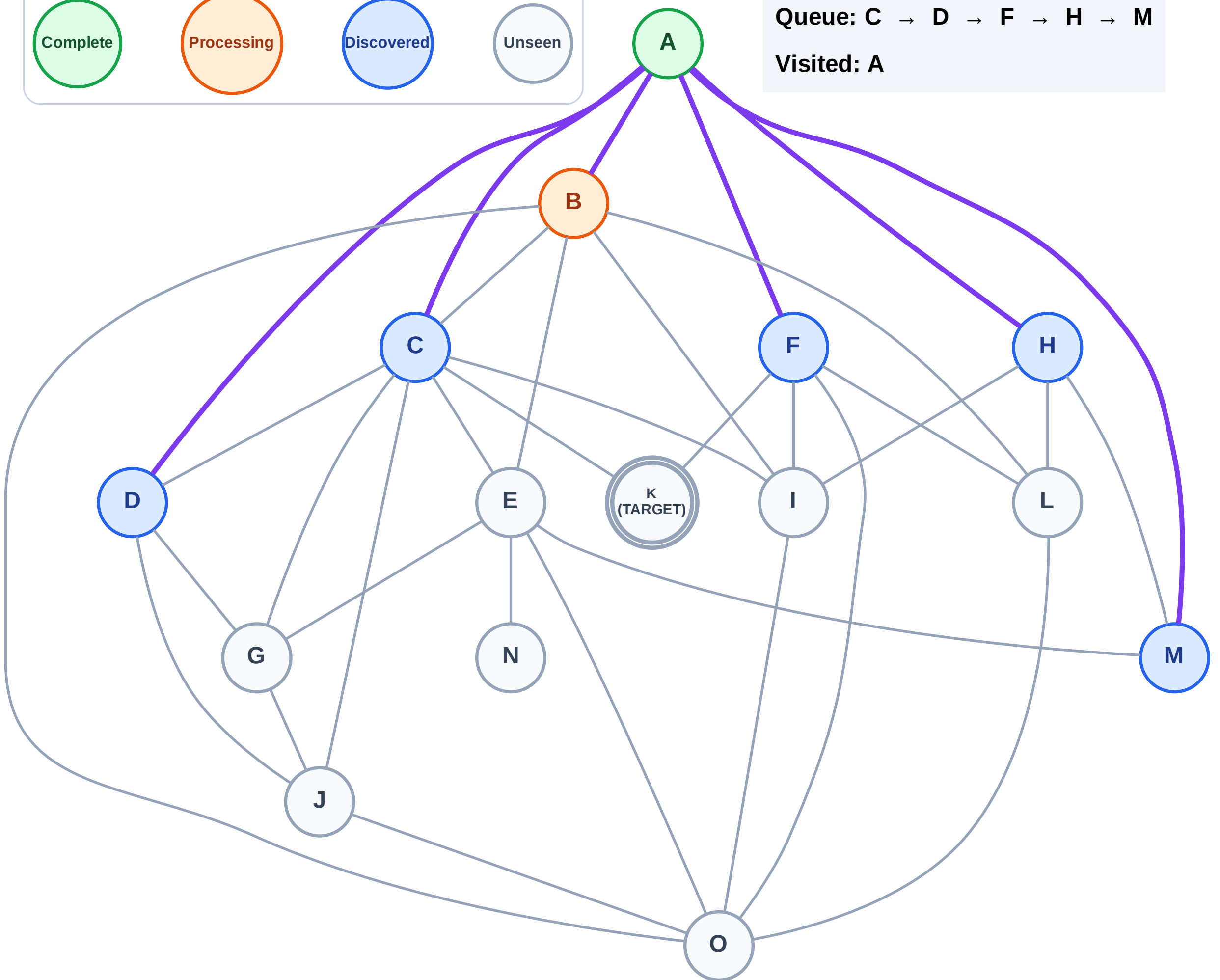
Step 4: Process node B

Legend



Queue: C → D → F → H → M

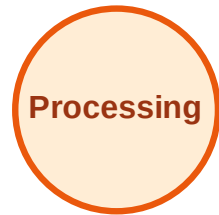
Visited: A



BFS Traversal

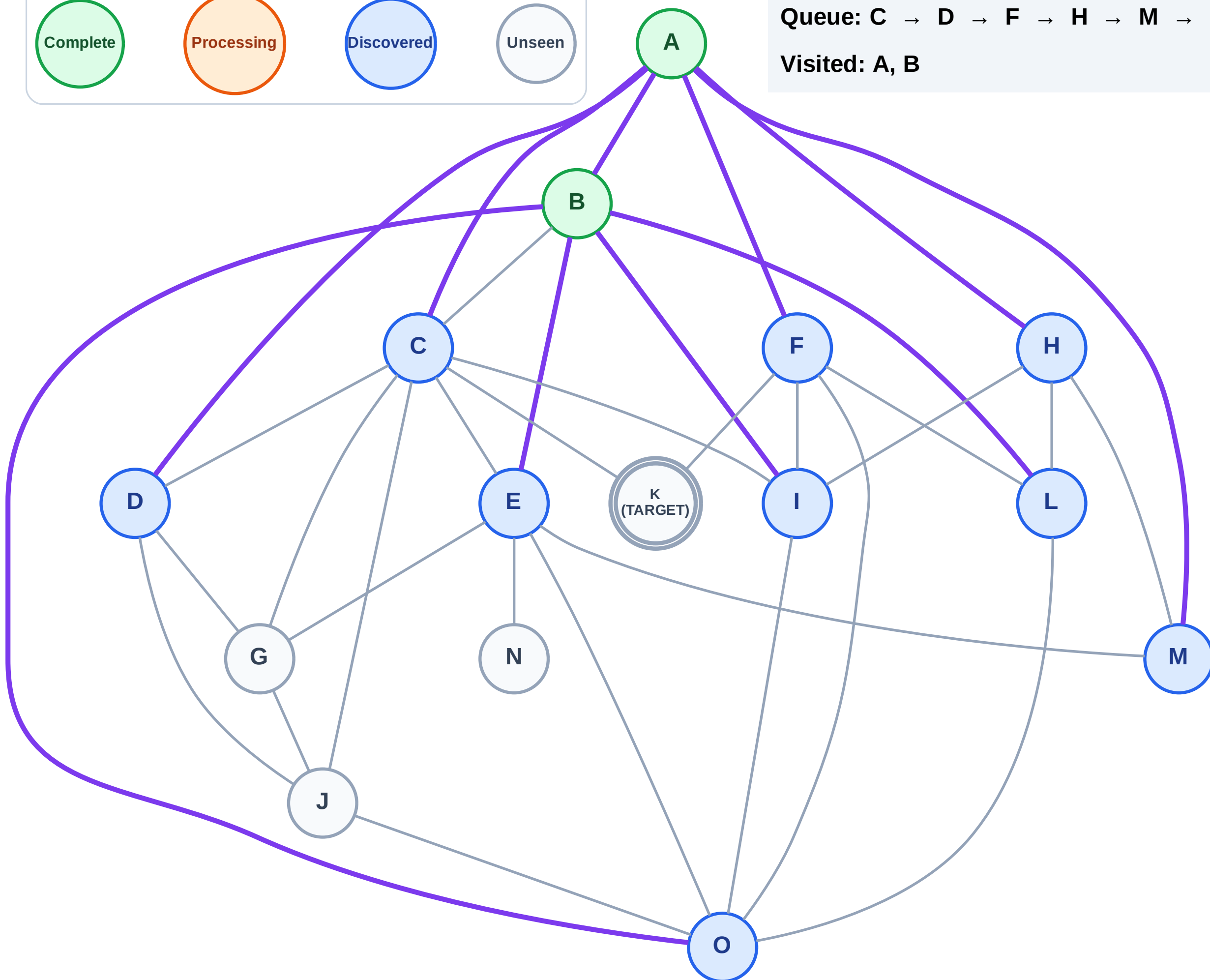
Step 5: Discover E, I, L, O from B

Legend



Queue: C → D → F → H → M → E → I → L → O

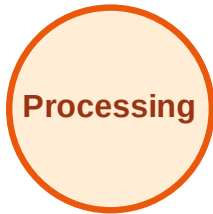
Visited: A, B



BFS Traversal

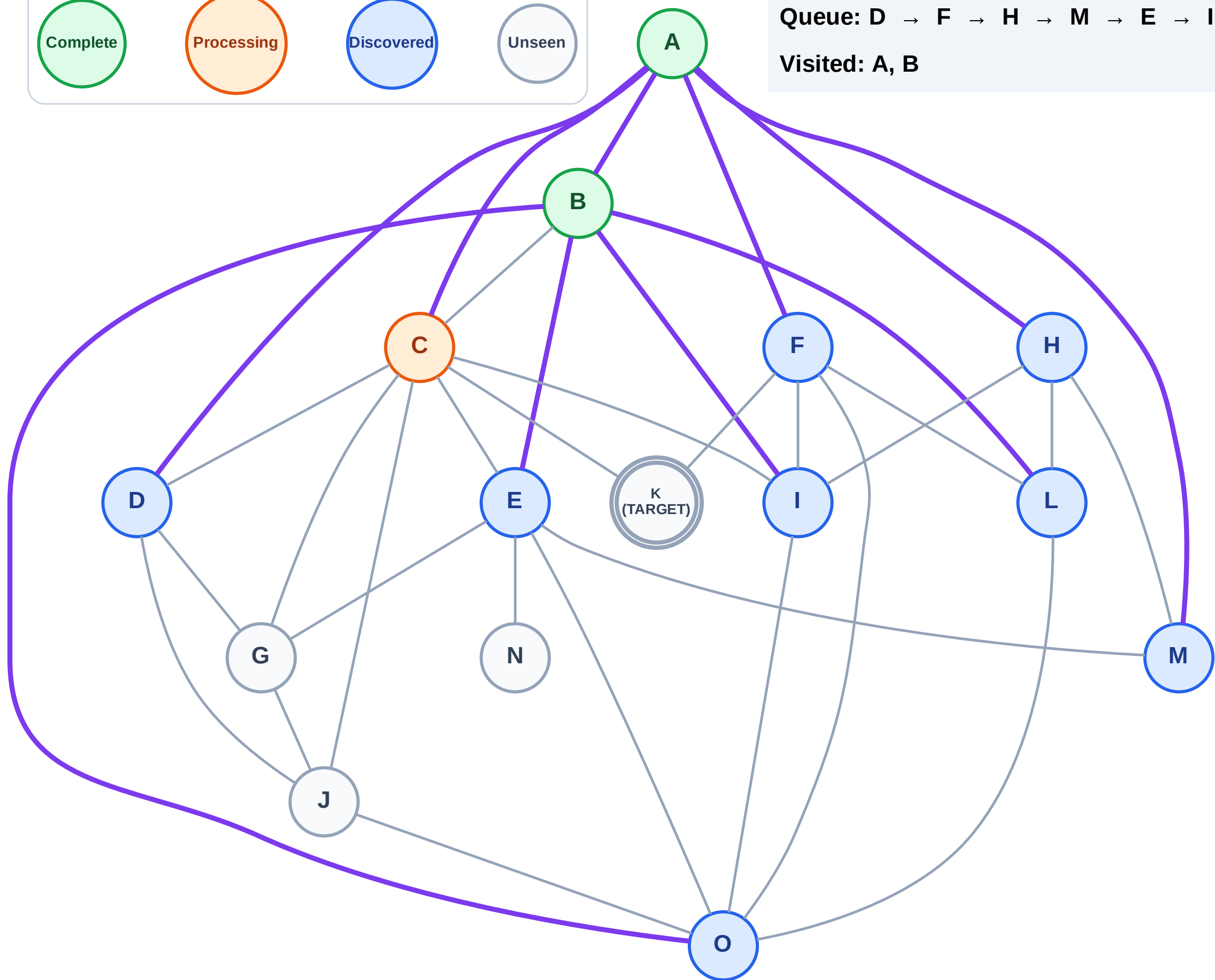
Step 6: Process node C

Legend



Queue: D → F → H → M → E → I → L → O

Visited: A, B



BFS Traversal

Step 7: Discover G, J, K from C

Legend

Complete

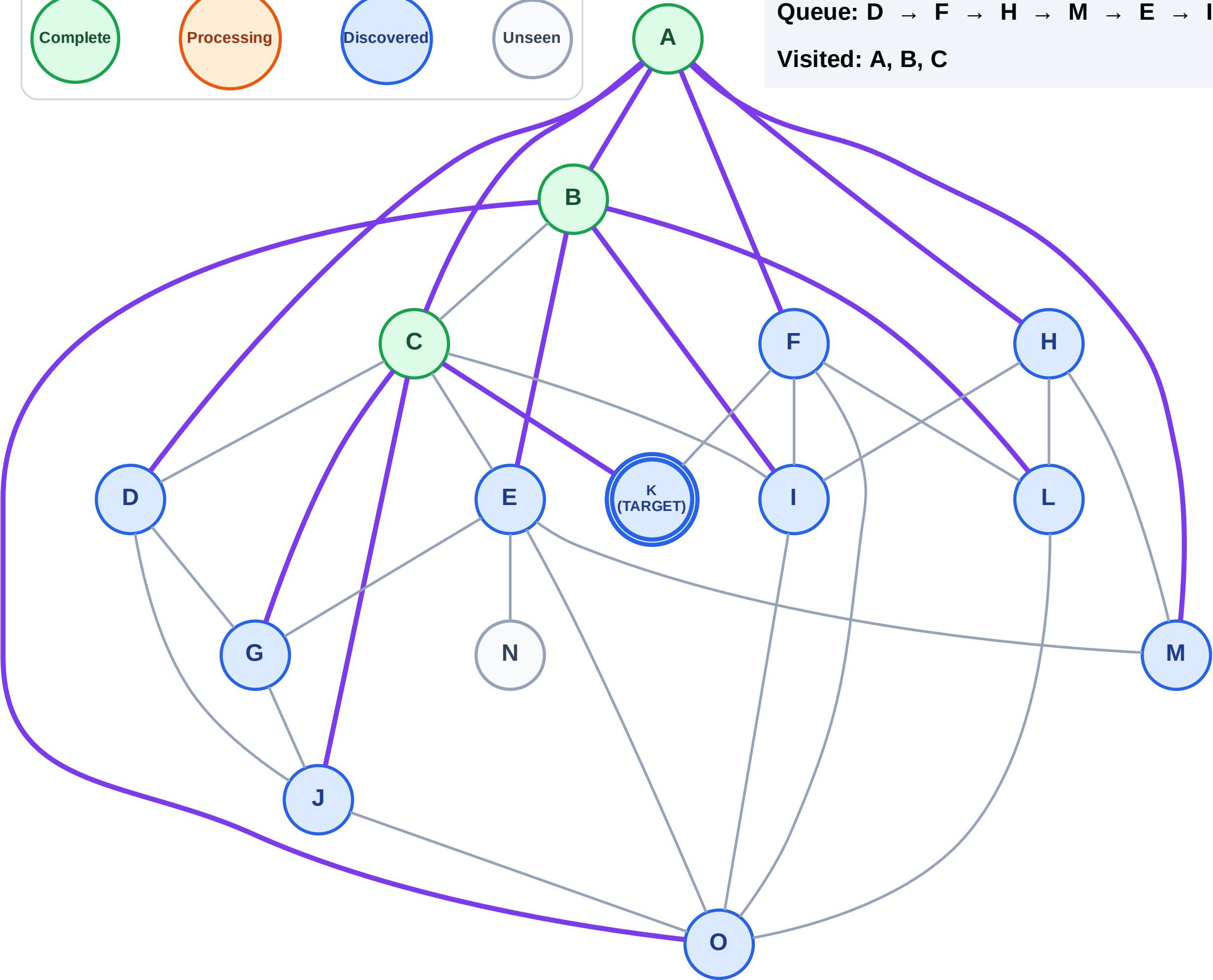
Processing

Discovered

Unseen

Queue: D → F → H → M → E → I → L → O → G → J → K

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

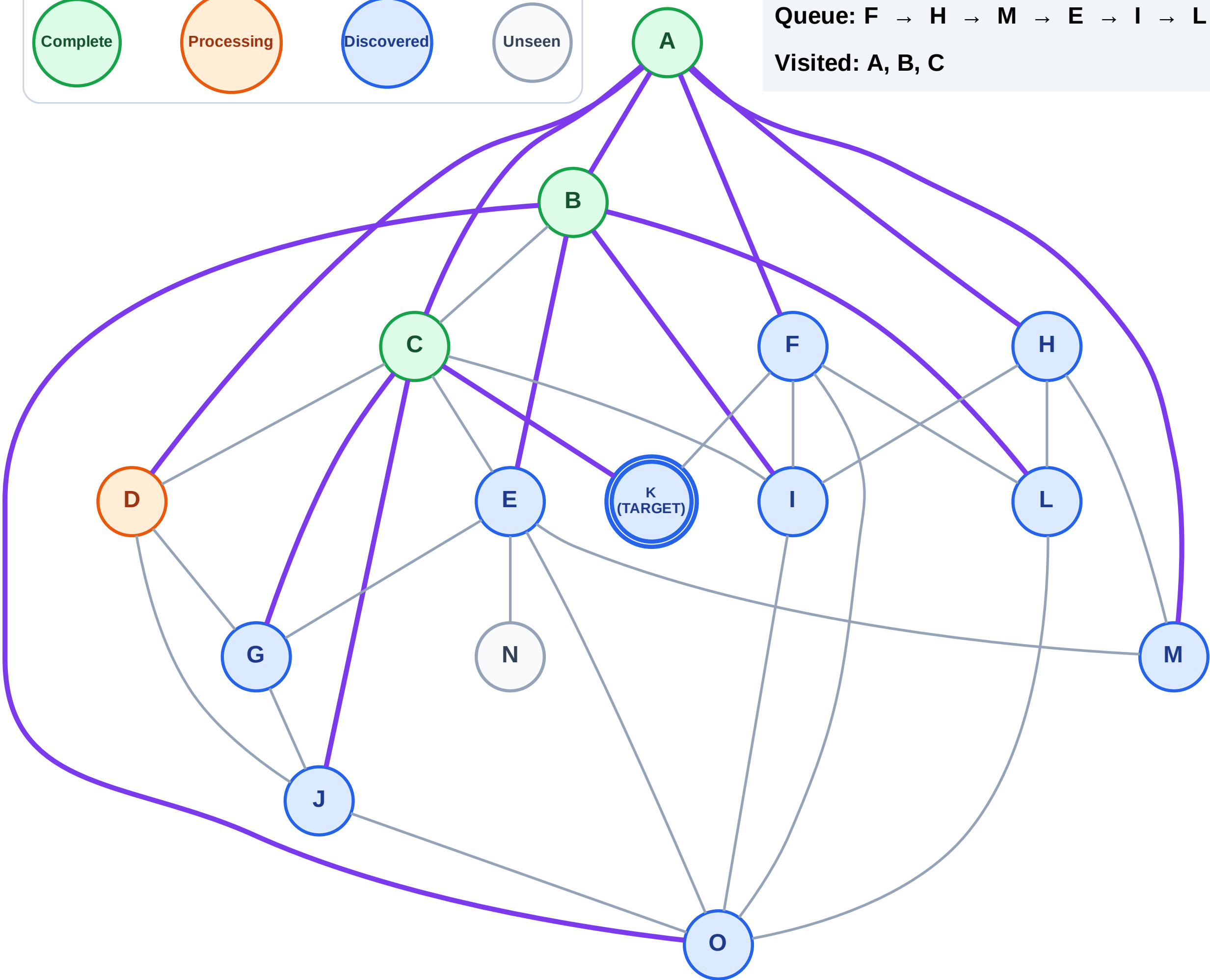
Processing

Discovered

Unseen

Queue: F → H → M → E → I → L → O → G → J → K

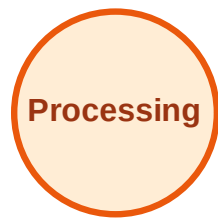
Visited: A, B, C



BFS Traversal

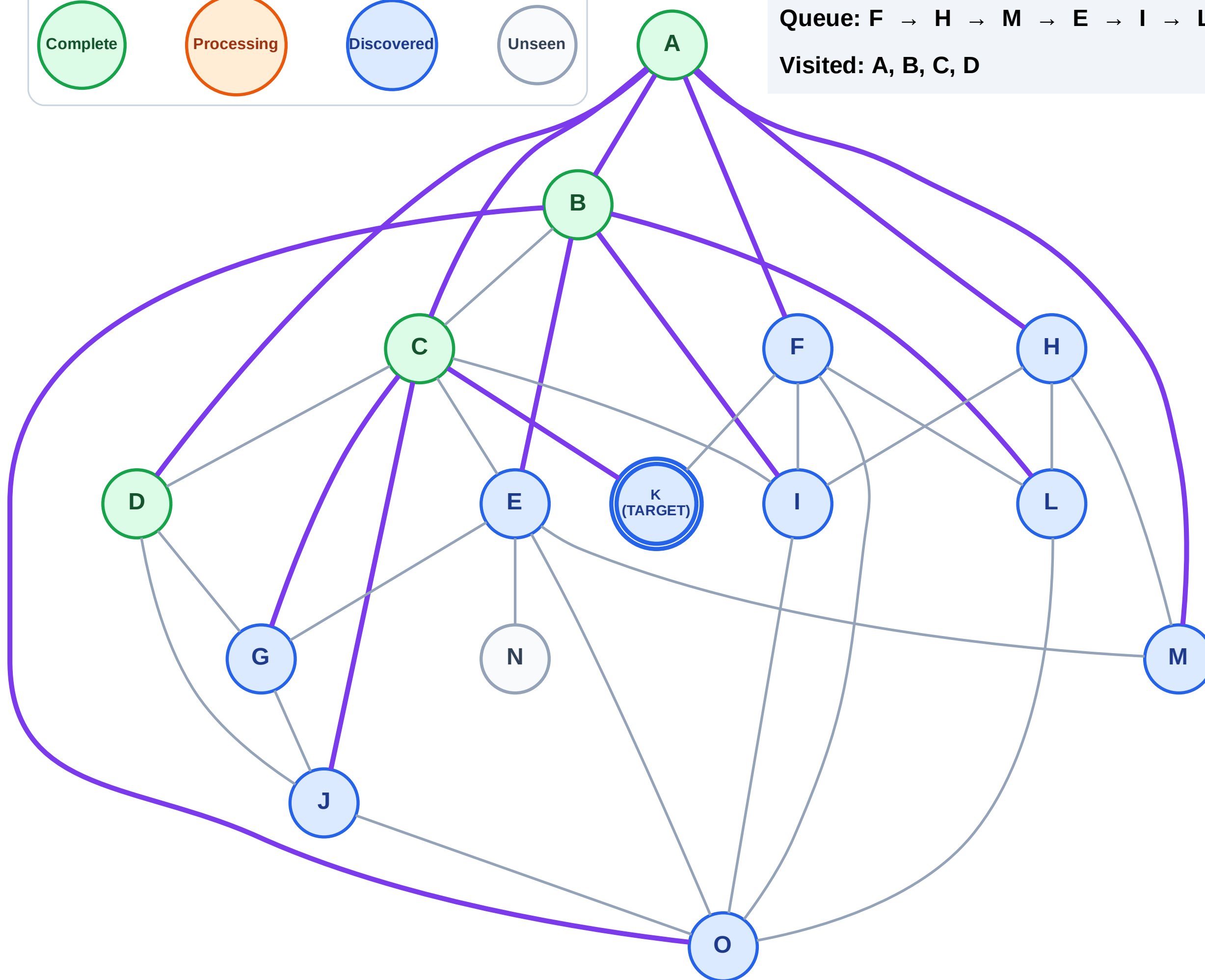
Step 9: No new neighbors from D

Legend



Queue: F → H → M → E → I → L → O → G → J → K

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

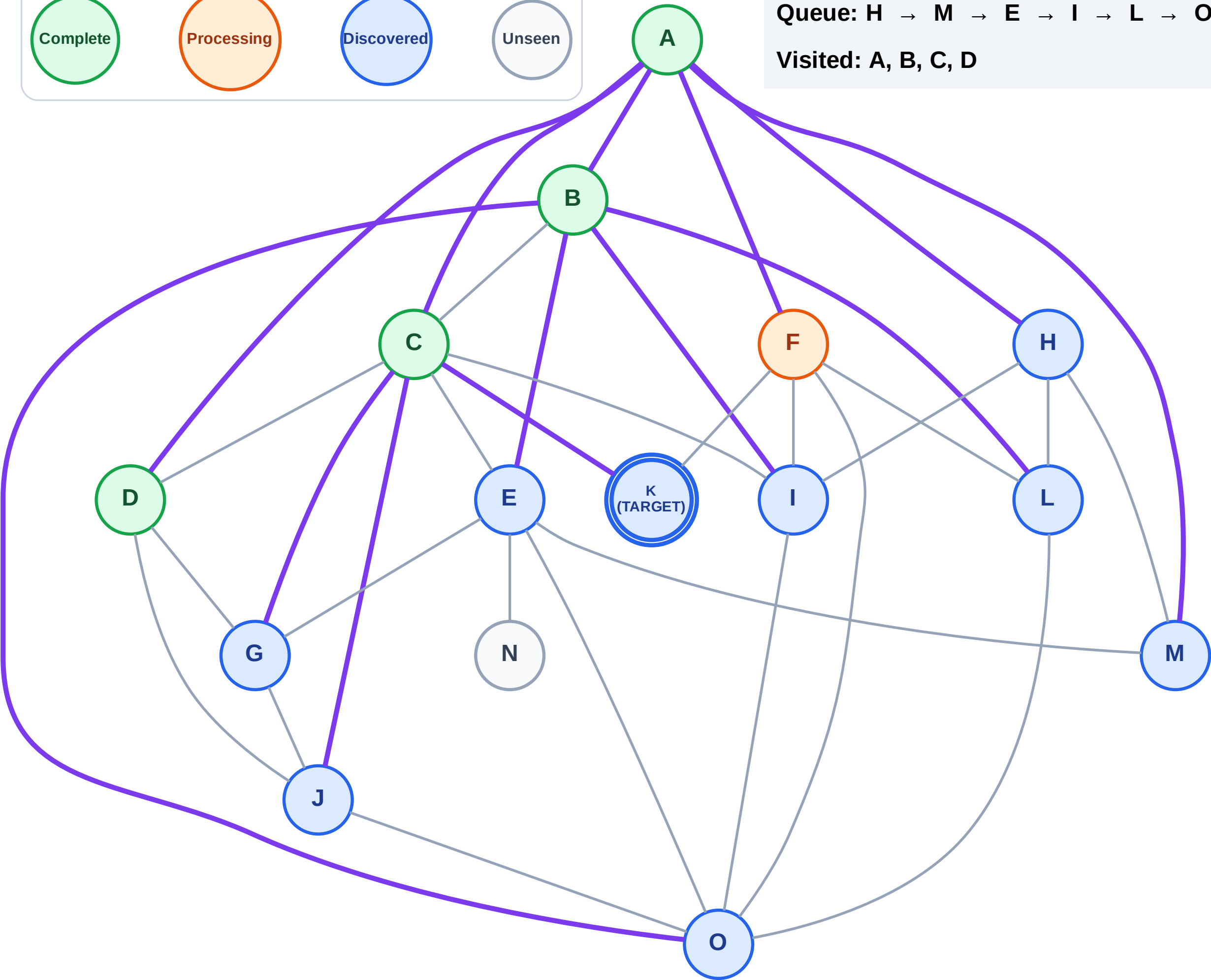
Processing

Discovered

Unseen

Queue: H → M → E → I → L → O → G → J → K

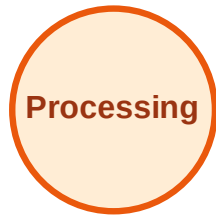
Visited: A, B, C, D



BFS Traversal

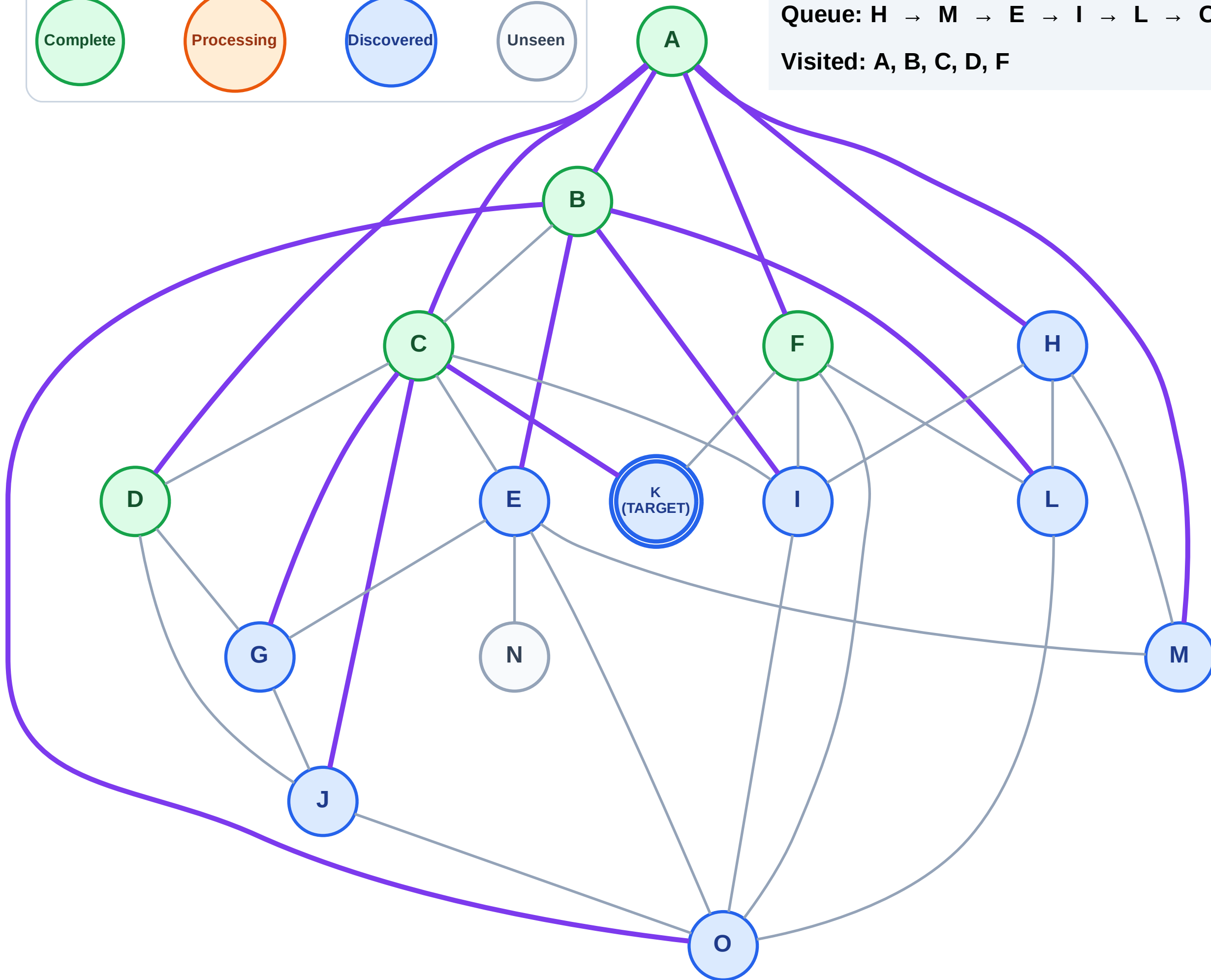
Step 11: No new neighbors from F

Legend



Queue: H → M → E → I → L → O → G → J → K

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node H

Legend

Complete

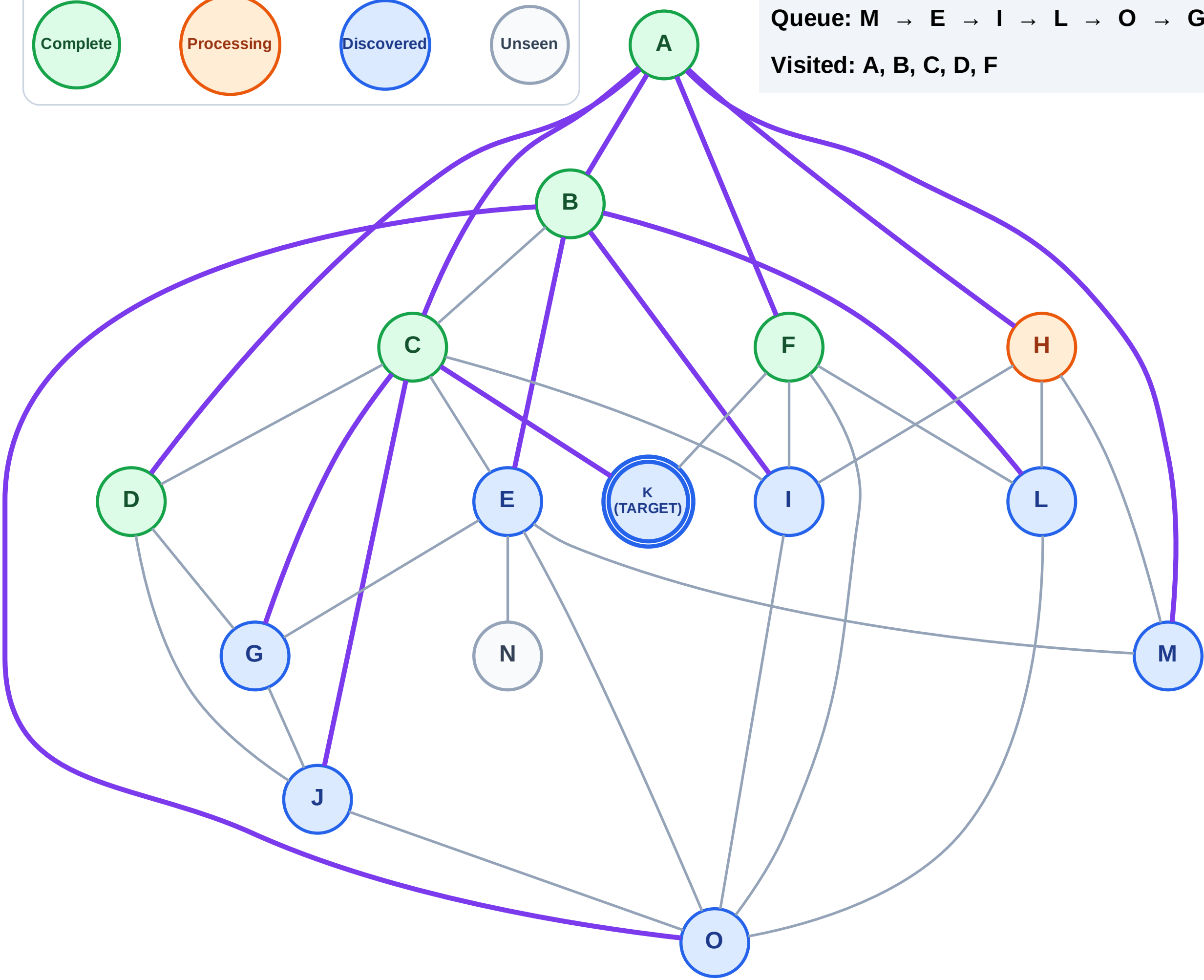
Processing

Discovered

Unseen

Queue: M → E → I → L → O → G → J → K

Visited: A, B, C, D, F

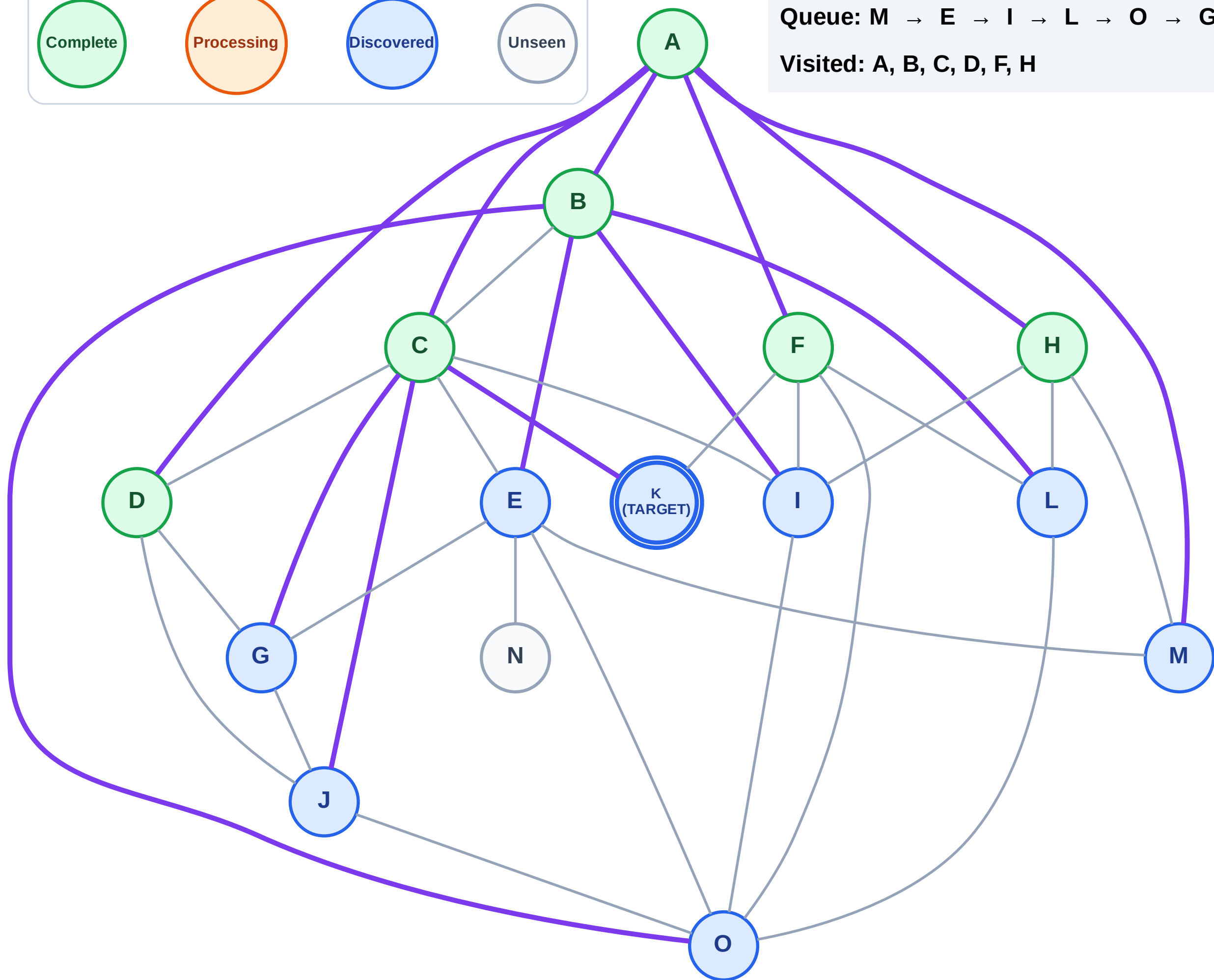


Step 13: No new neighbors from H

Legend

Unseen

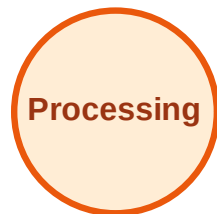
Visited: A, B, C, D, F, H



BFS Traversal

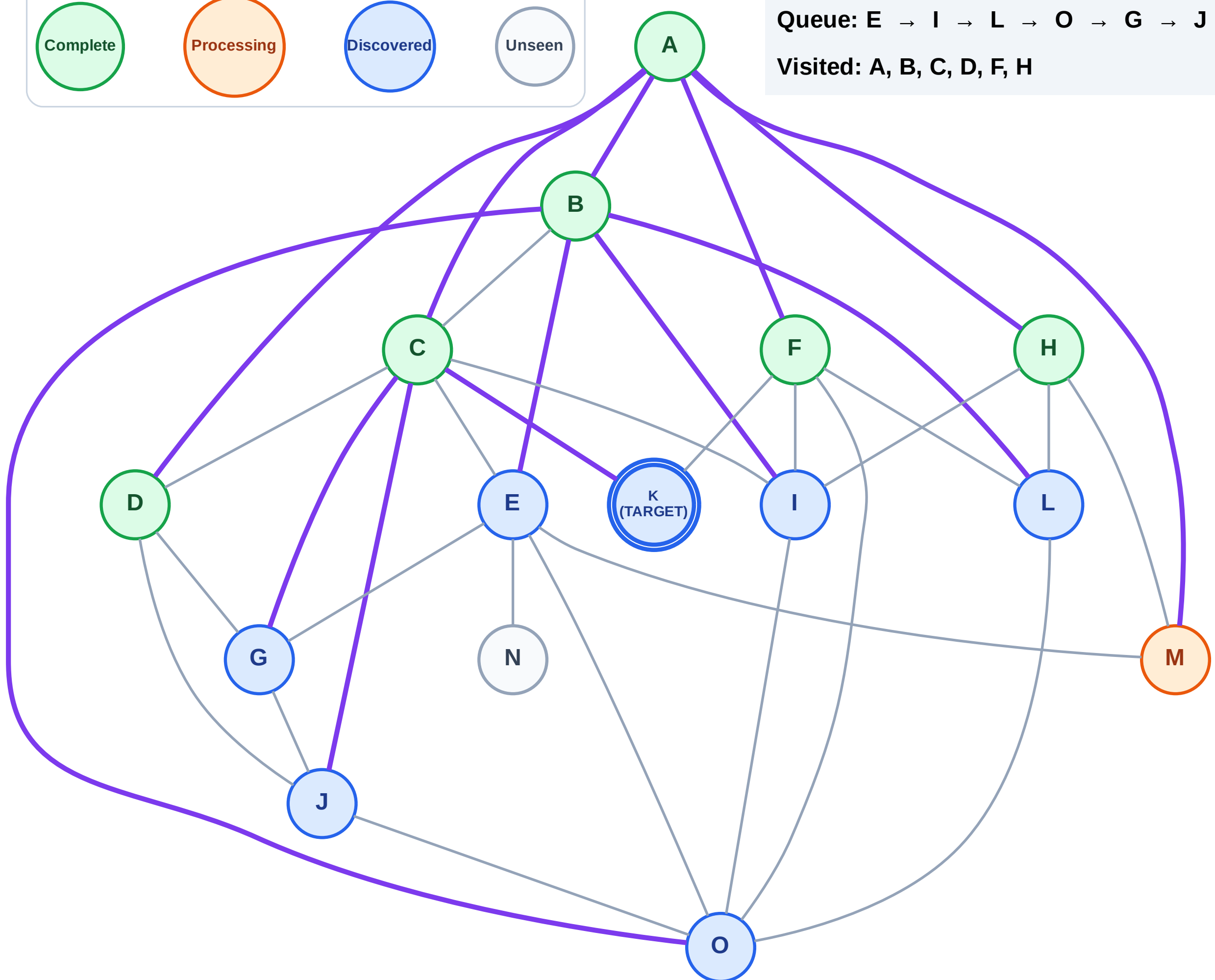
Step 14: Process node M

Legend



Queue: E → I → L → O → G → J → K

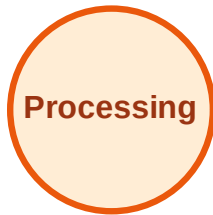
Visited: A, B, C, D, F, H



BFS Traversal

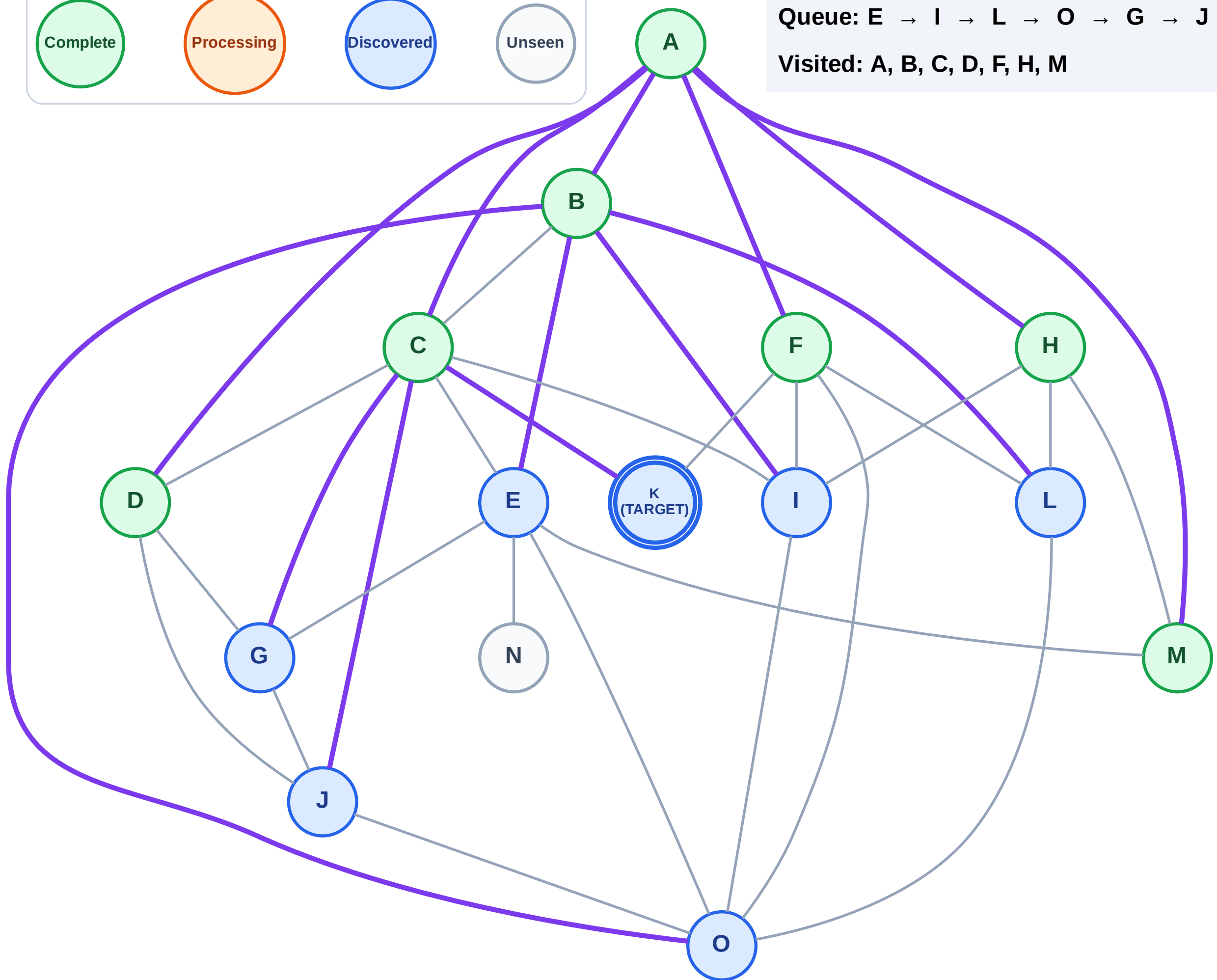
Step 15: No new neighbors from M

Legend



Queue: E → I → L → O → G → J → K

Visited: A, B, C, D, F, H, M



BFS Traversal

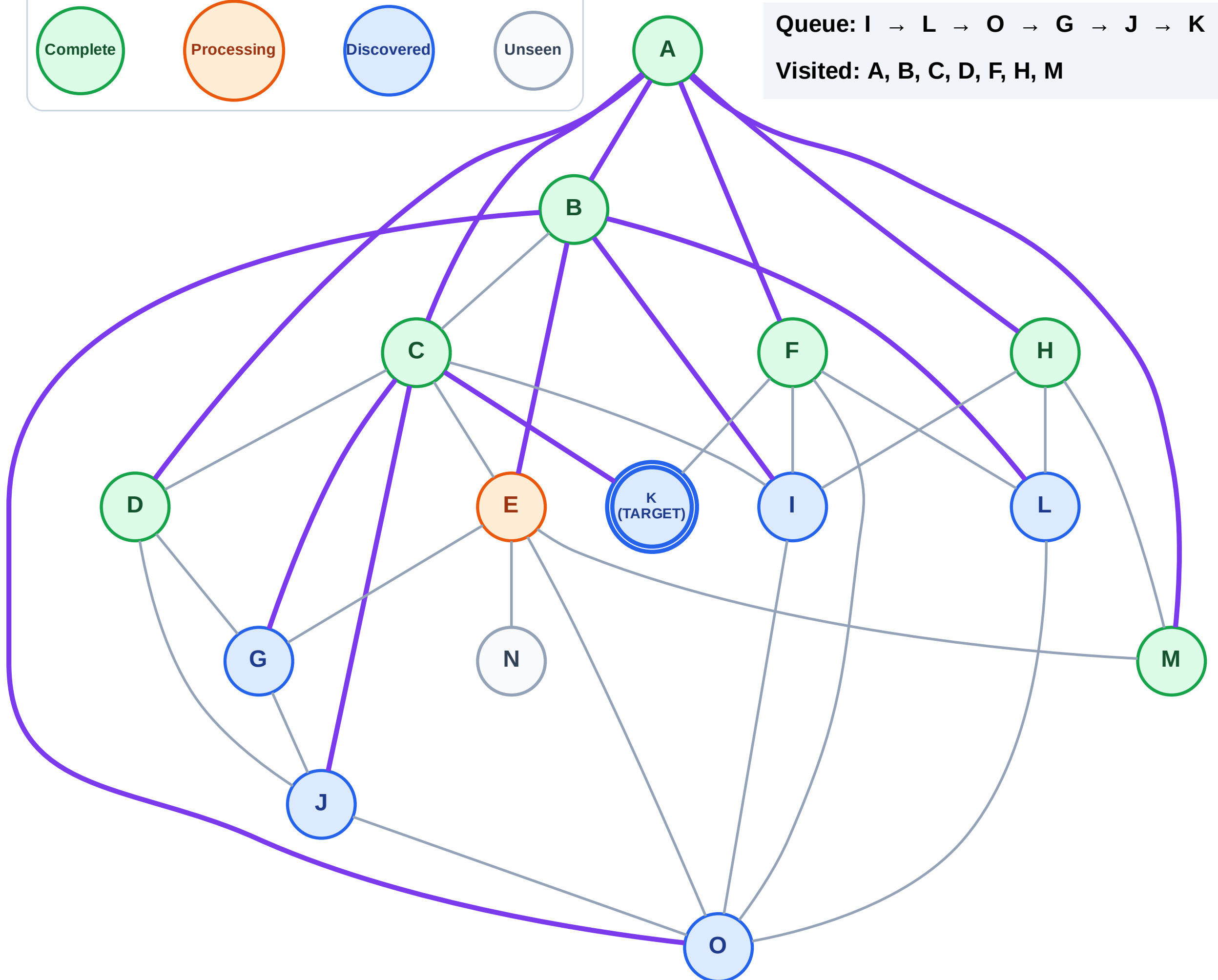
Step 16: Process node E

Legend



Queue: I → L → O → G → J → K

Visited: A, B, C, D, F, H, M



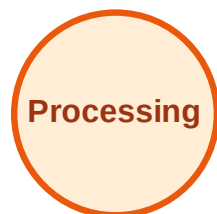
BFS Traversal

Step 17: Discover N from E

Legend



Complete



Processing



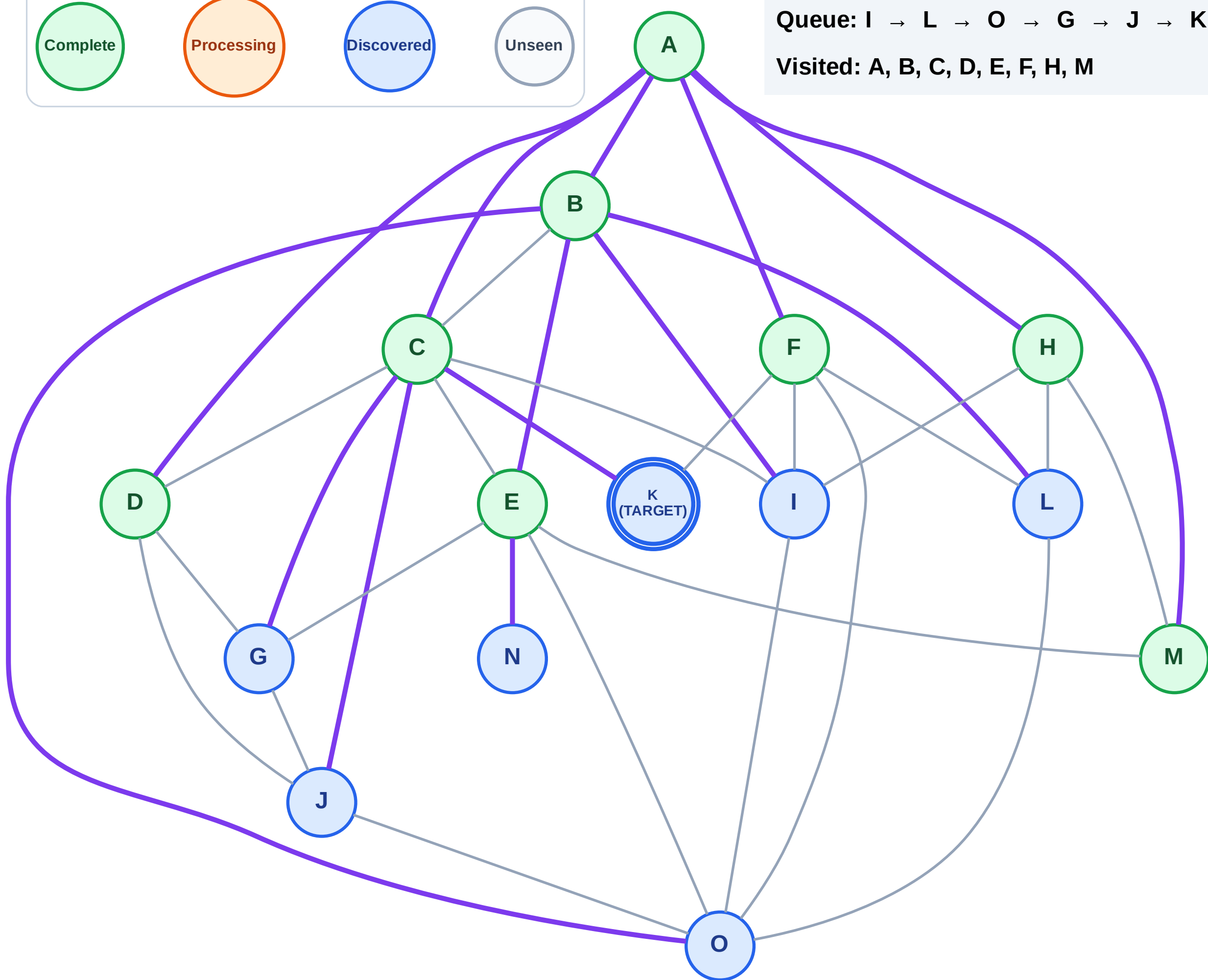
Discovered



Unseen

Queue: I → L → O → G → J → K → N

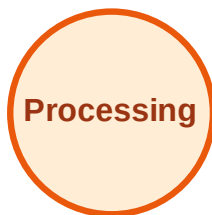
Visited: A, B, C, D, E, F, H, M



BFS Traversal

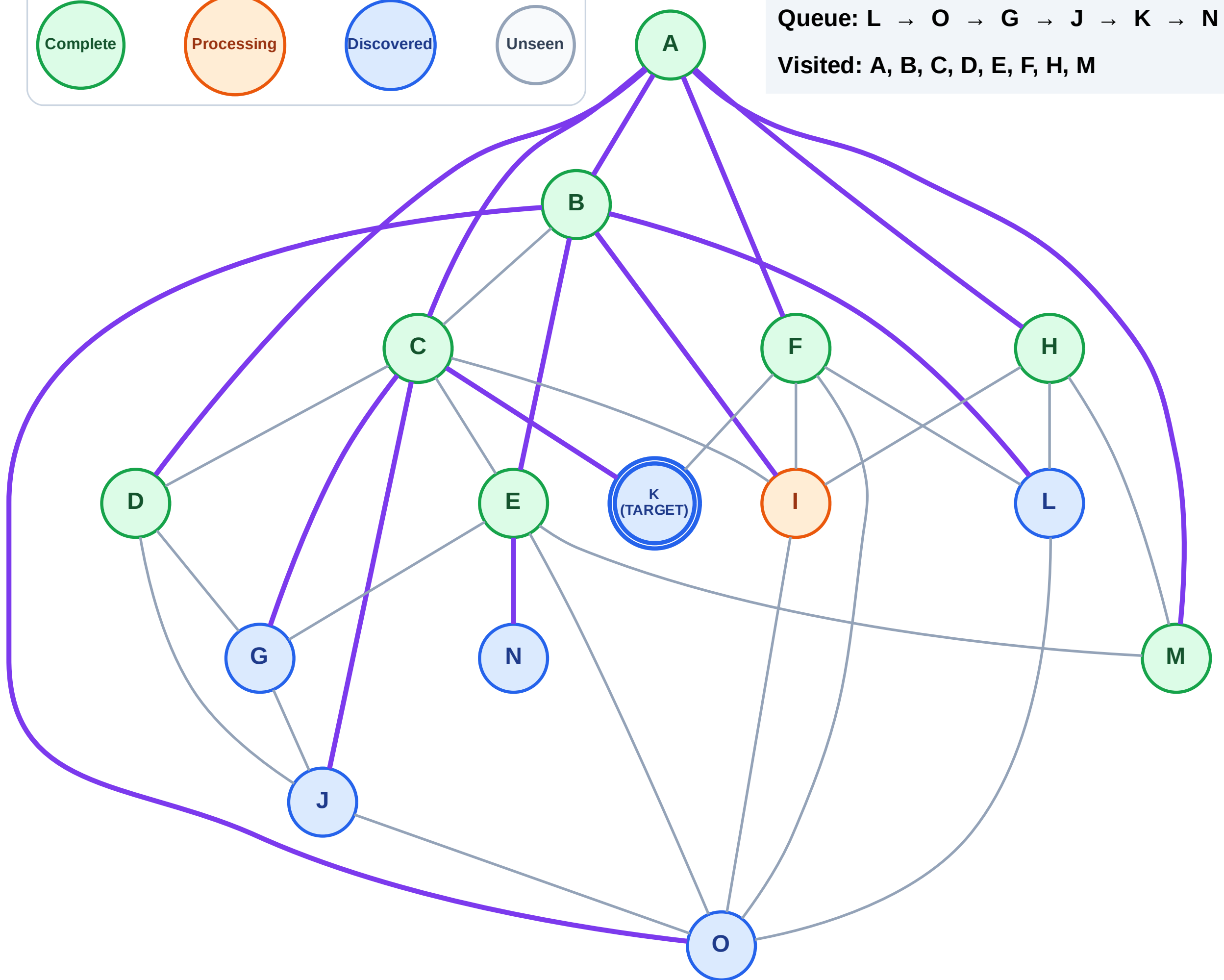
Step 18: Process node I

Legend



Queue: L → O → G → J → K → N

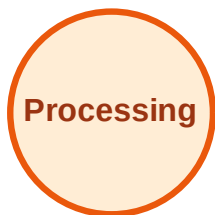
Visited: A, B, C, D, E, F, H, M



BFS Traversal

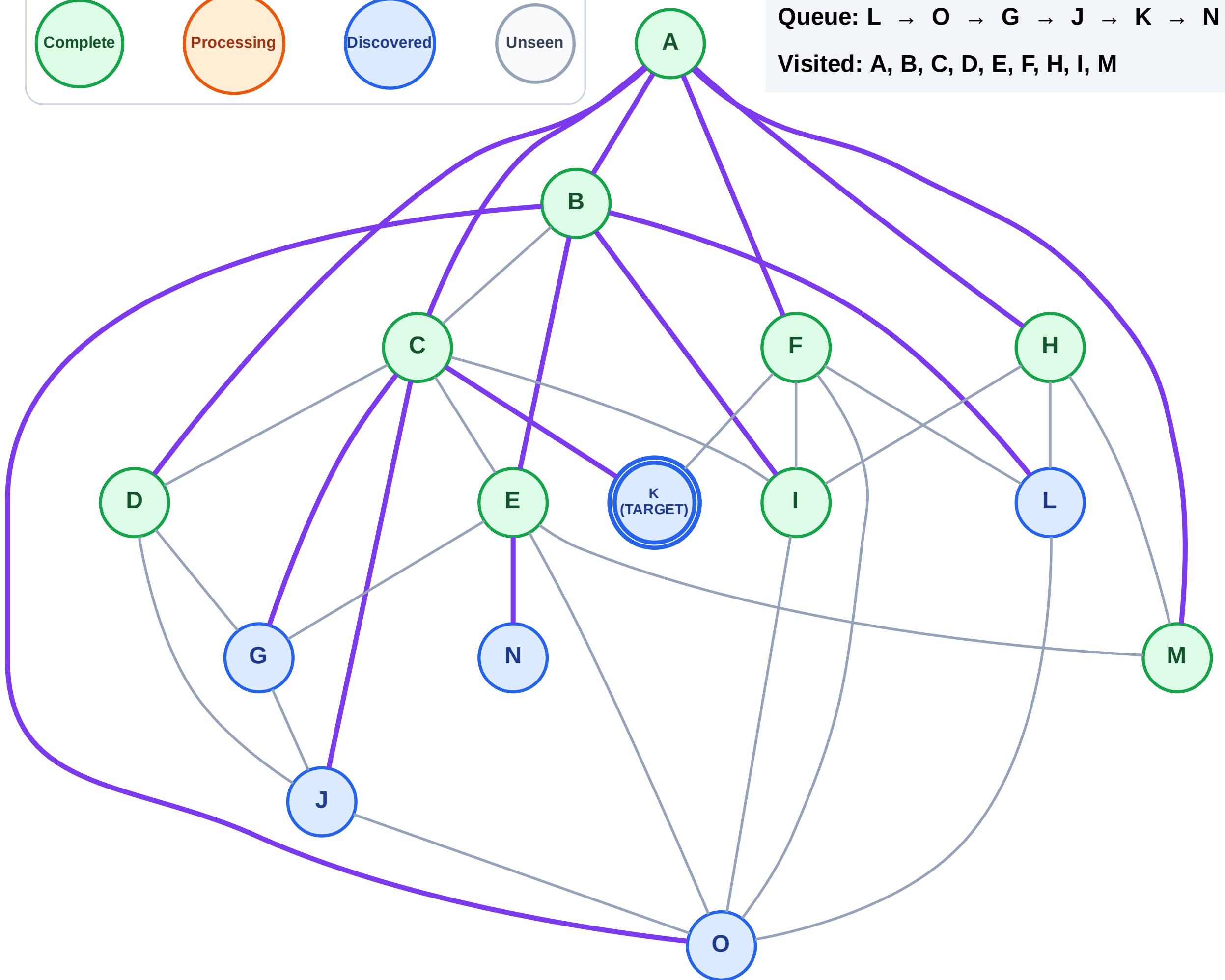
Step 19: No new neighbors from I

Legend



Queue: L → O → G → J → K → N

Visited: A, B, C, D, E, F, H, I, M



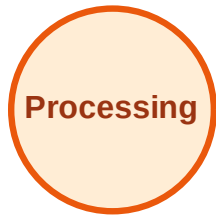
BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



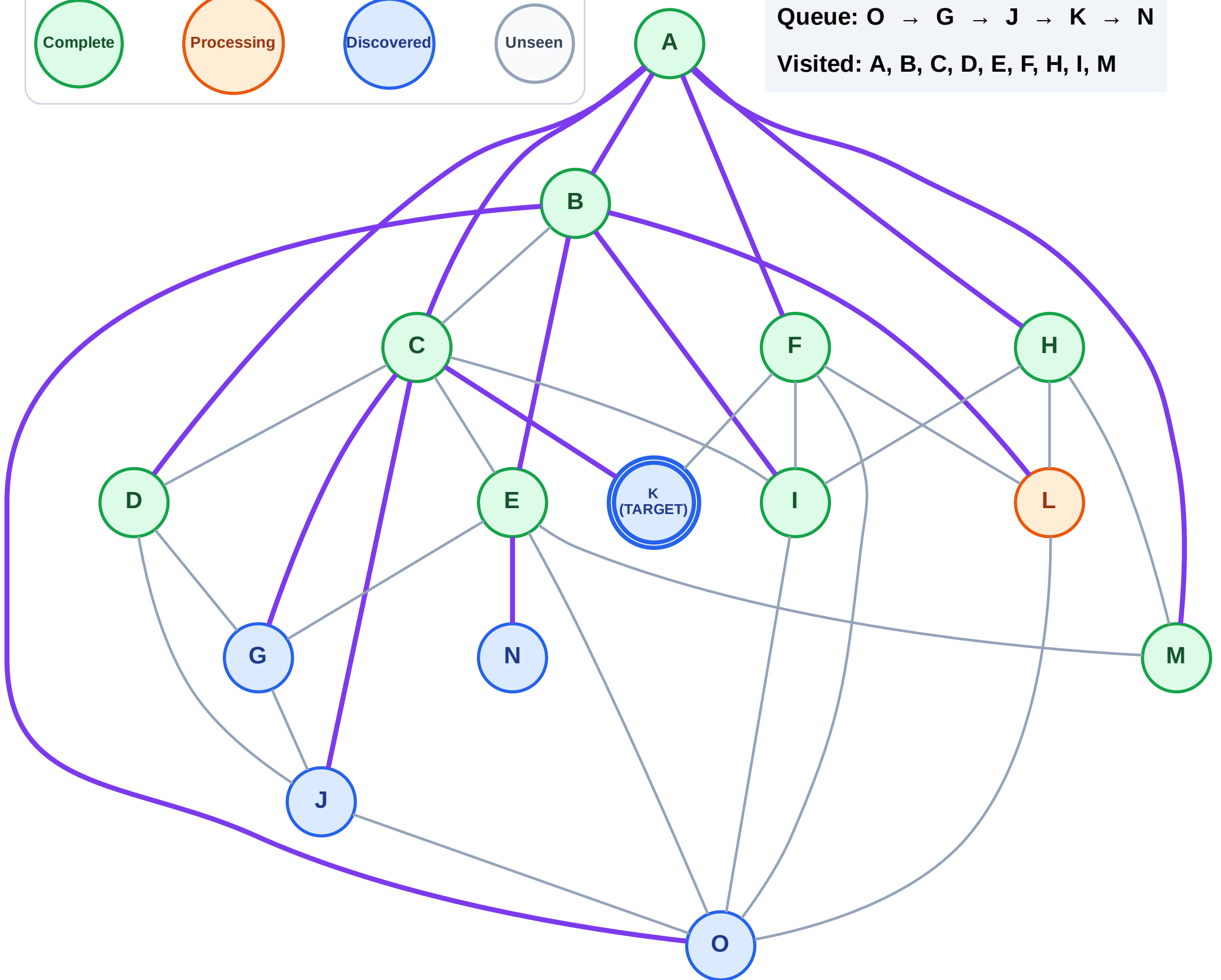
Discovered



Unseen

Queue: O → G → J → K → N

Visited: A, B, C, D, E, F, H, I, M



BFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



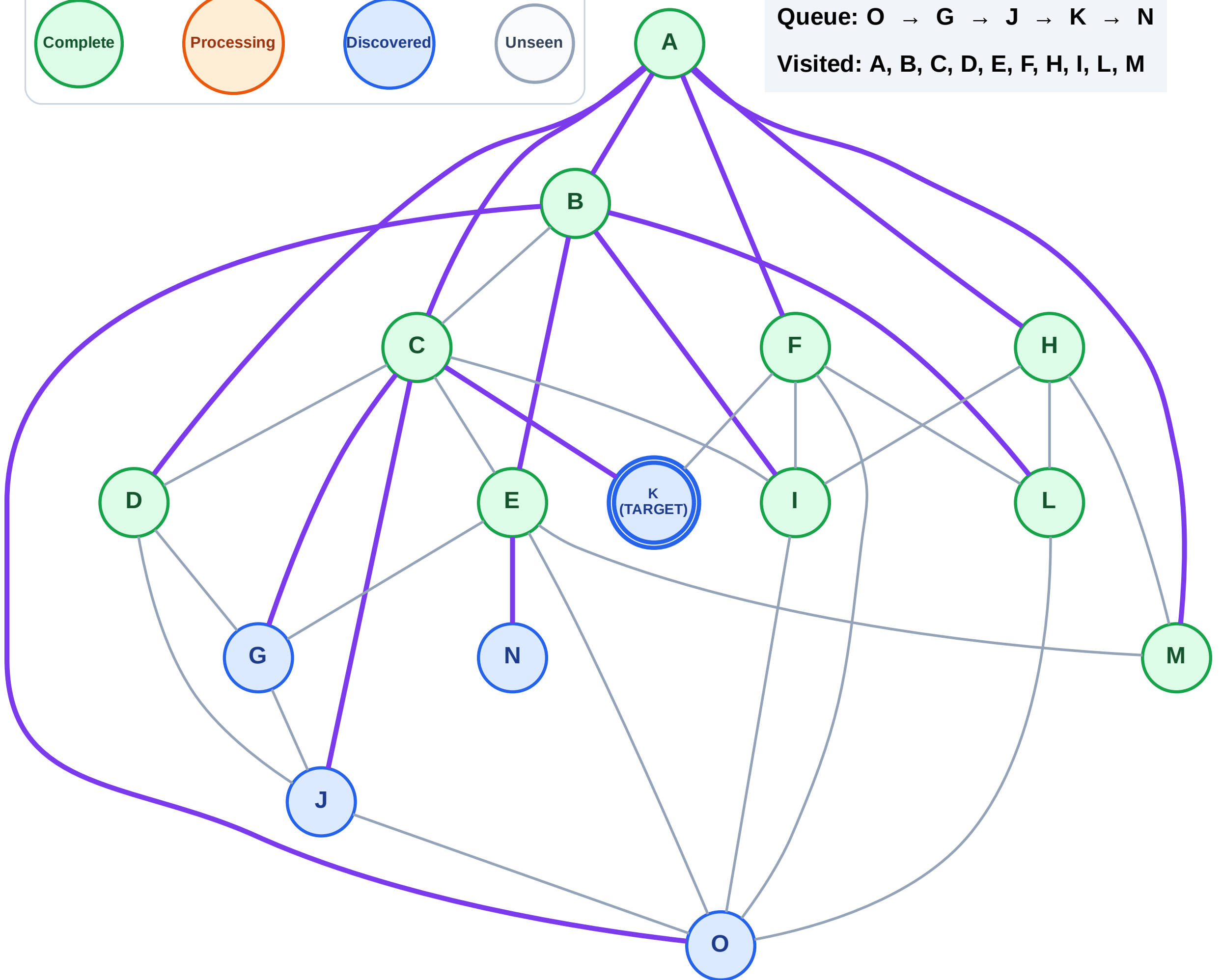
Discovered



Unseen

Queue: O → G → J → K → N

Visited: A, B, C, D, E, F, H, I, L, M



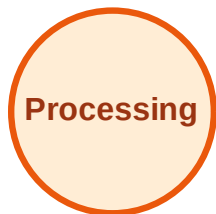
BFS Traversal

Step 22: Process node O

Legend



Complete



Processing



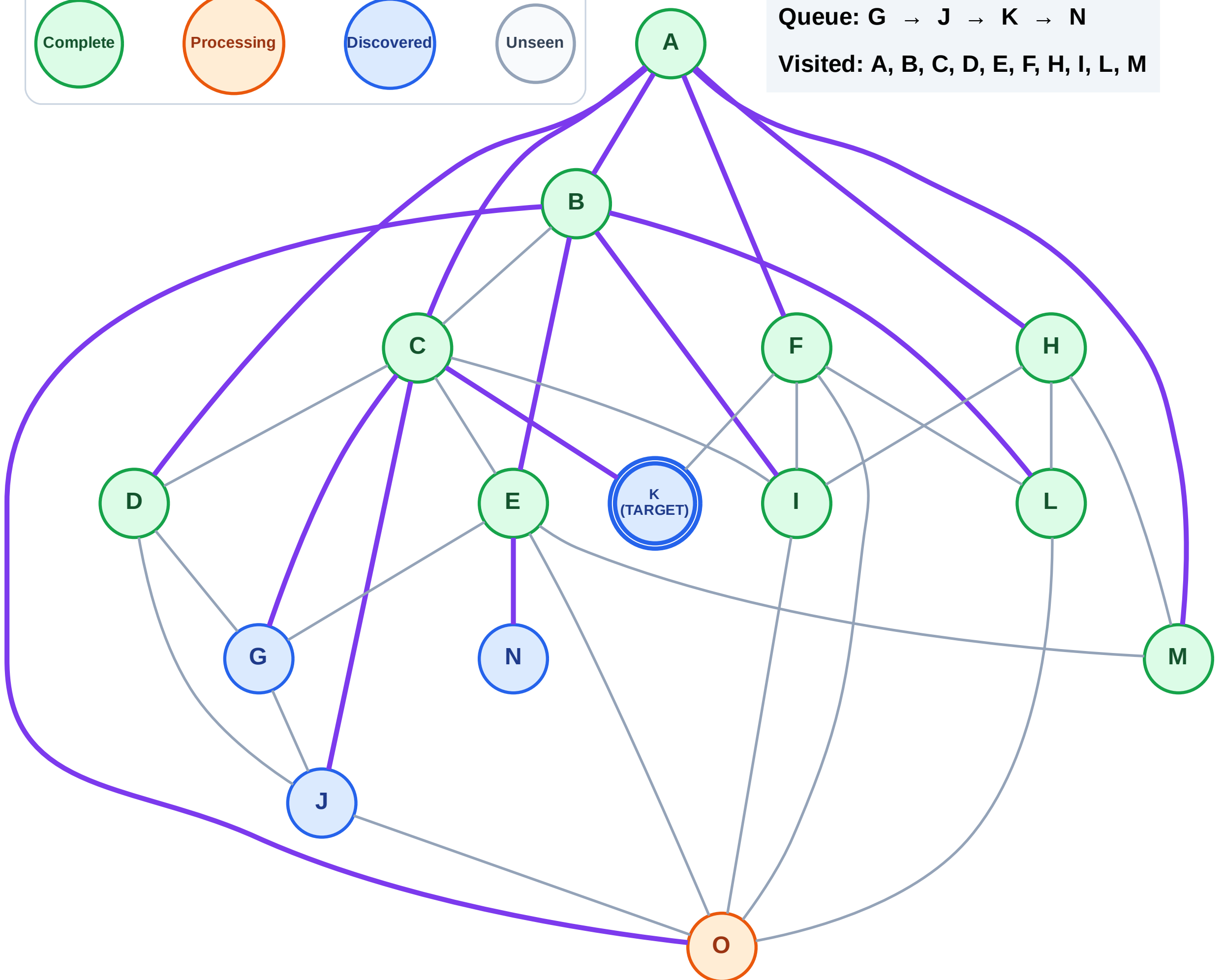
Discovered



Unseen

Queue: G → J → K → N

Visited: A, B, C, D, E, F, H, I, L, M



BFS Traversal

Step 23: No new neighbors from O

Legend



Complete



Processing



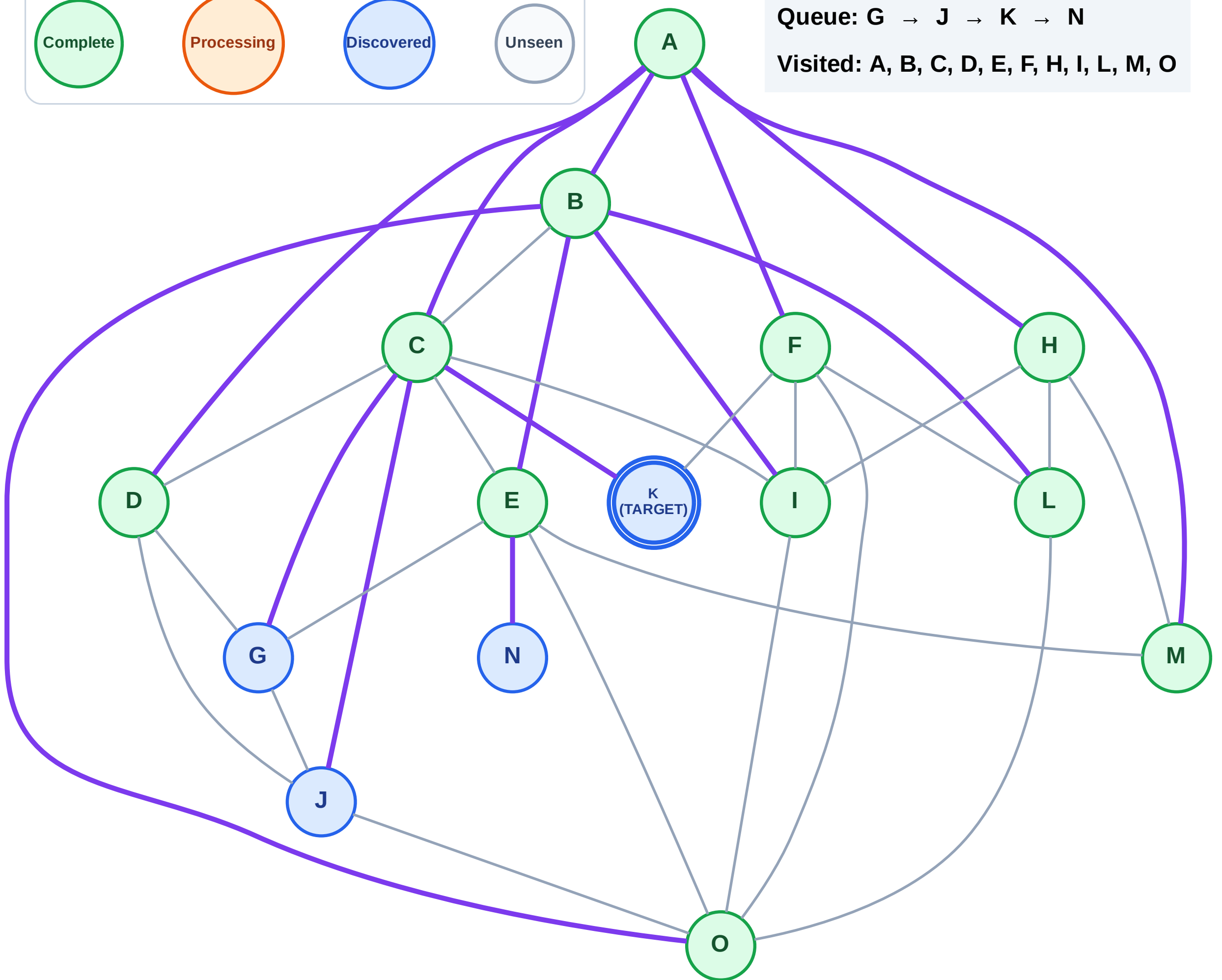
Discovered



Unseen

Queue: G → J → K → N

Visited: A, B, C, D, E, F, H, I, L, M, O



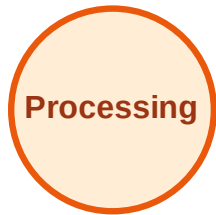
BFS Traversal

Step 24: Process node G

Legend



Complete



Processing



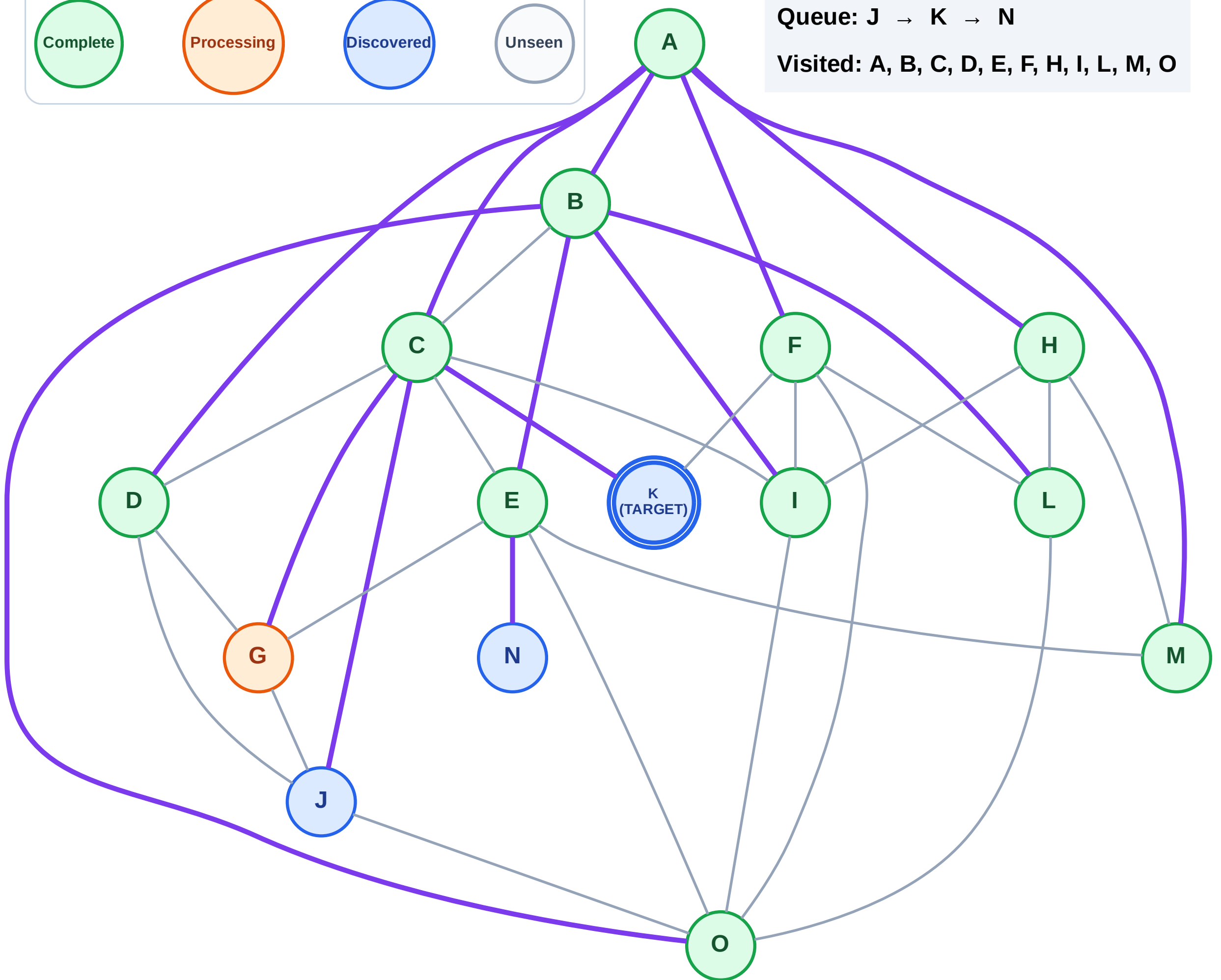
Discovered



Unseen

Queue: J → K → N

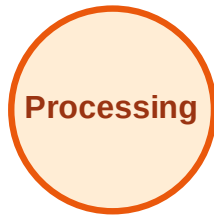
Visited: A, B, C, D, E, F, H, I, L, M, O



BFS Traversal

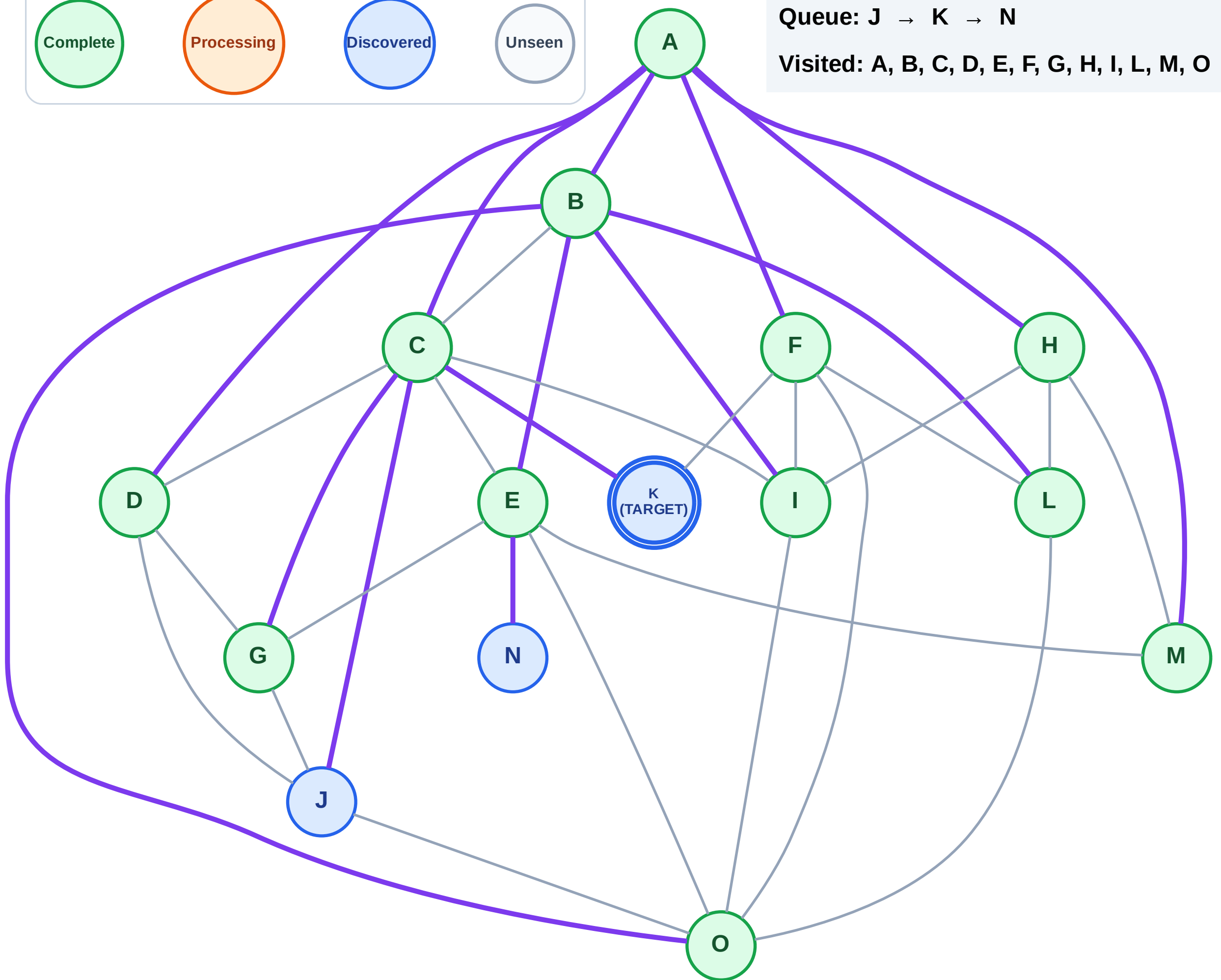
Step 25: No new neighbors from G

Legend



Queue: J → K → N

Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

Step 26: Process node J

Legend

Complete

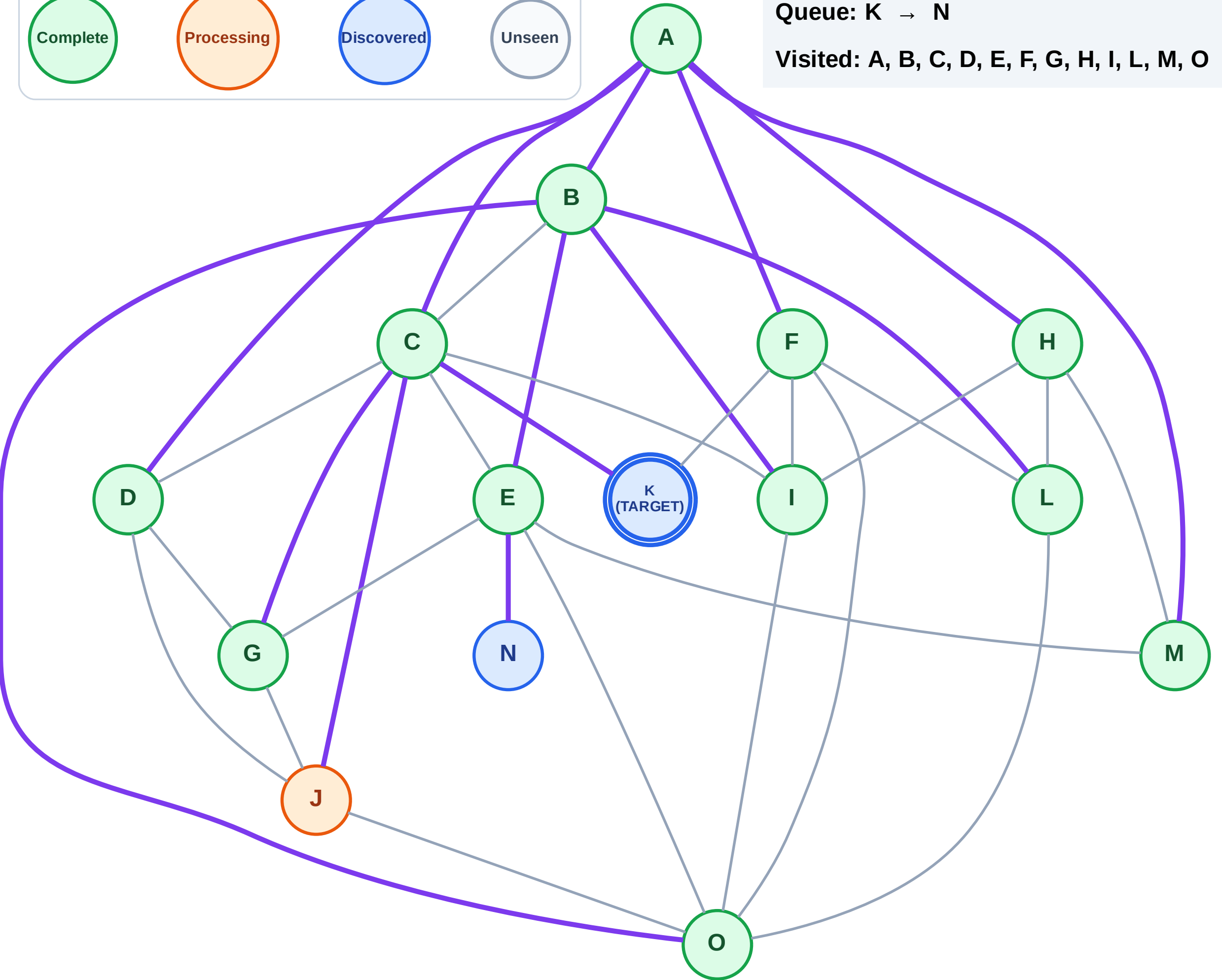
Processing

Discovered

Unseen

Queue: K → N

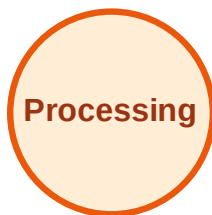
Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

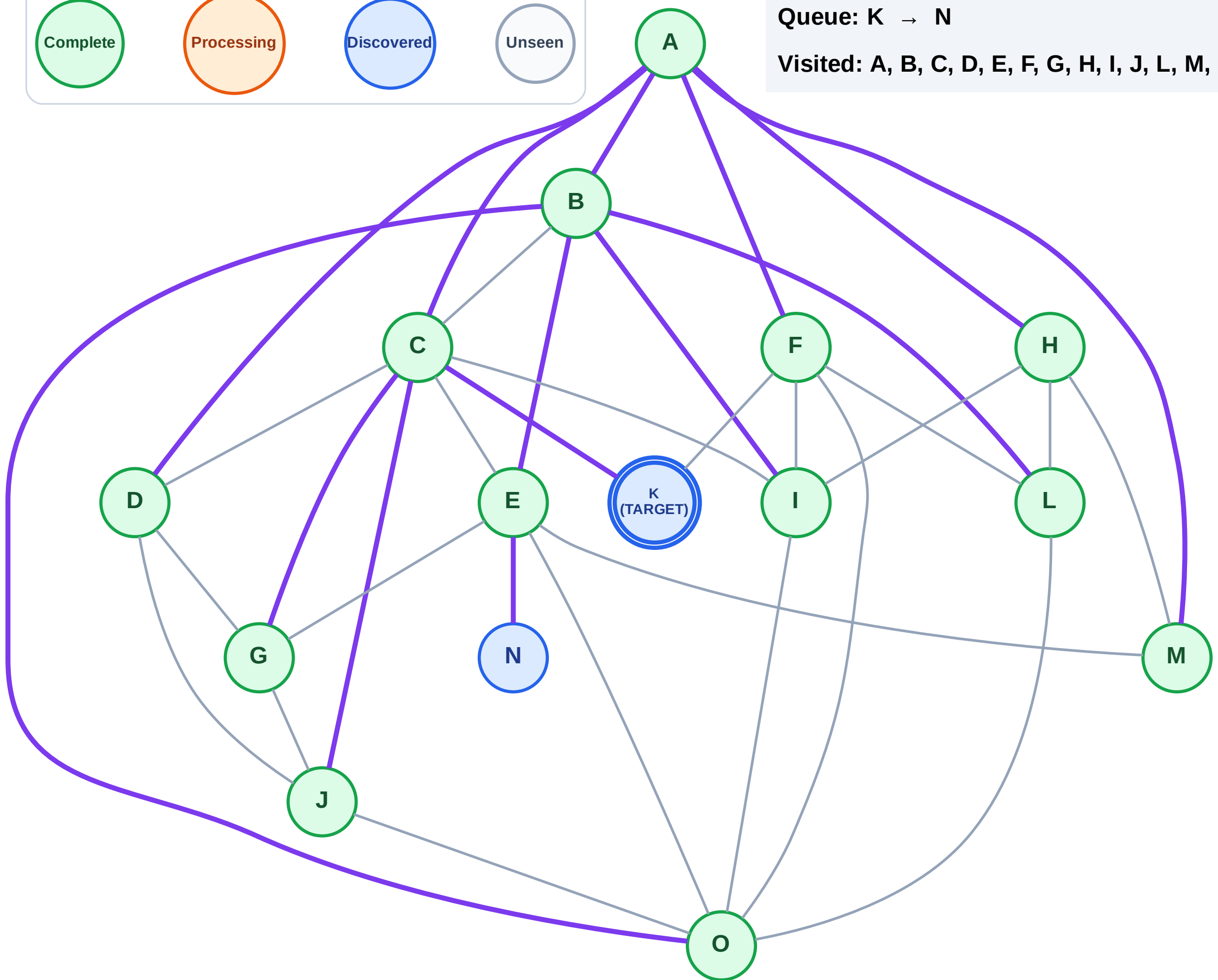
Step 27: No new neighbors from J

Legend



Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 28: Process node K

Legend

Complete

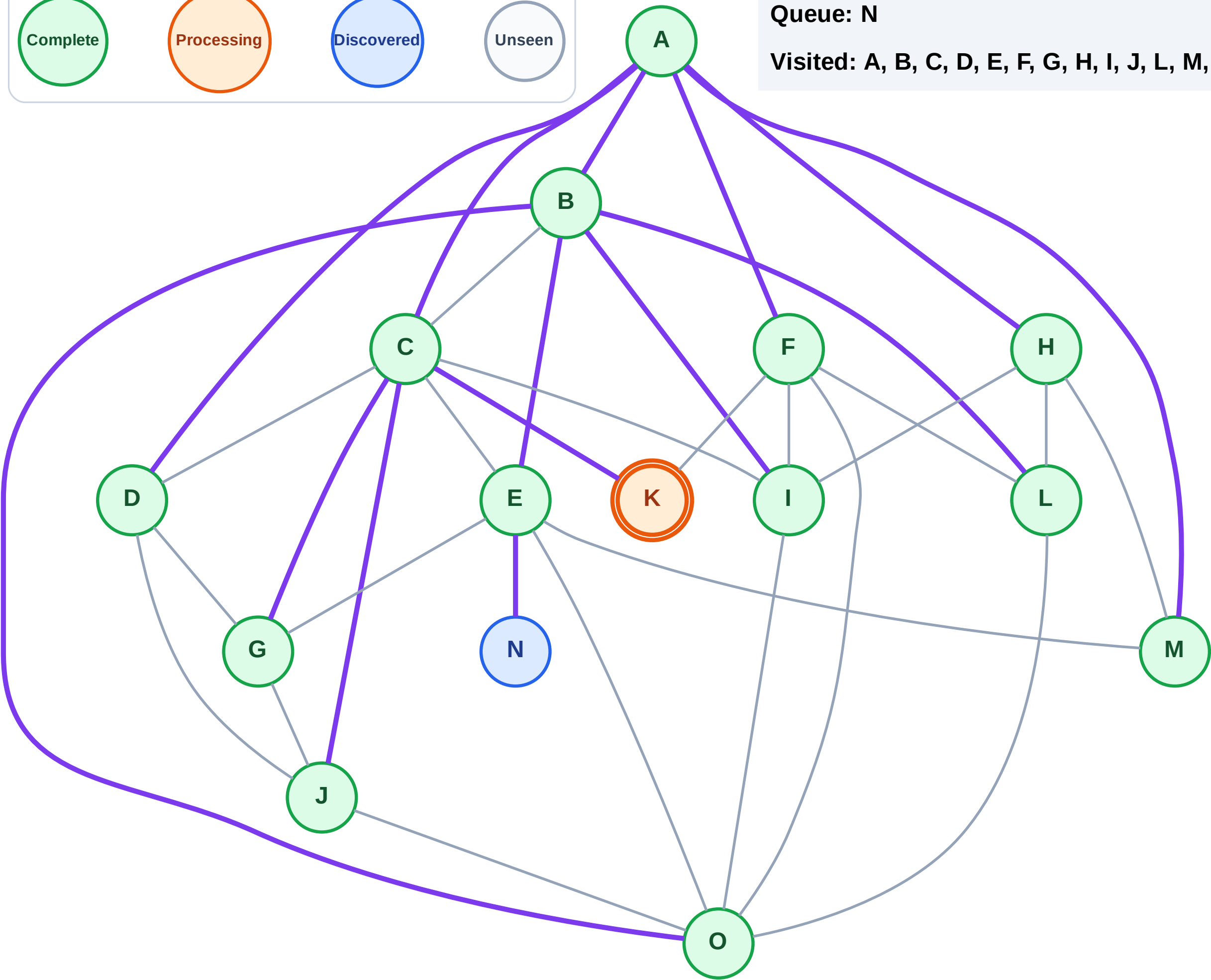
Processing

Discovered

Unseen

Queue: N

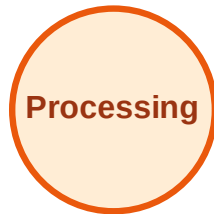
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

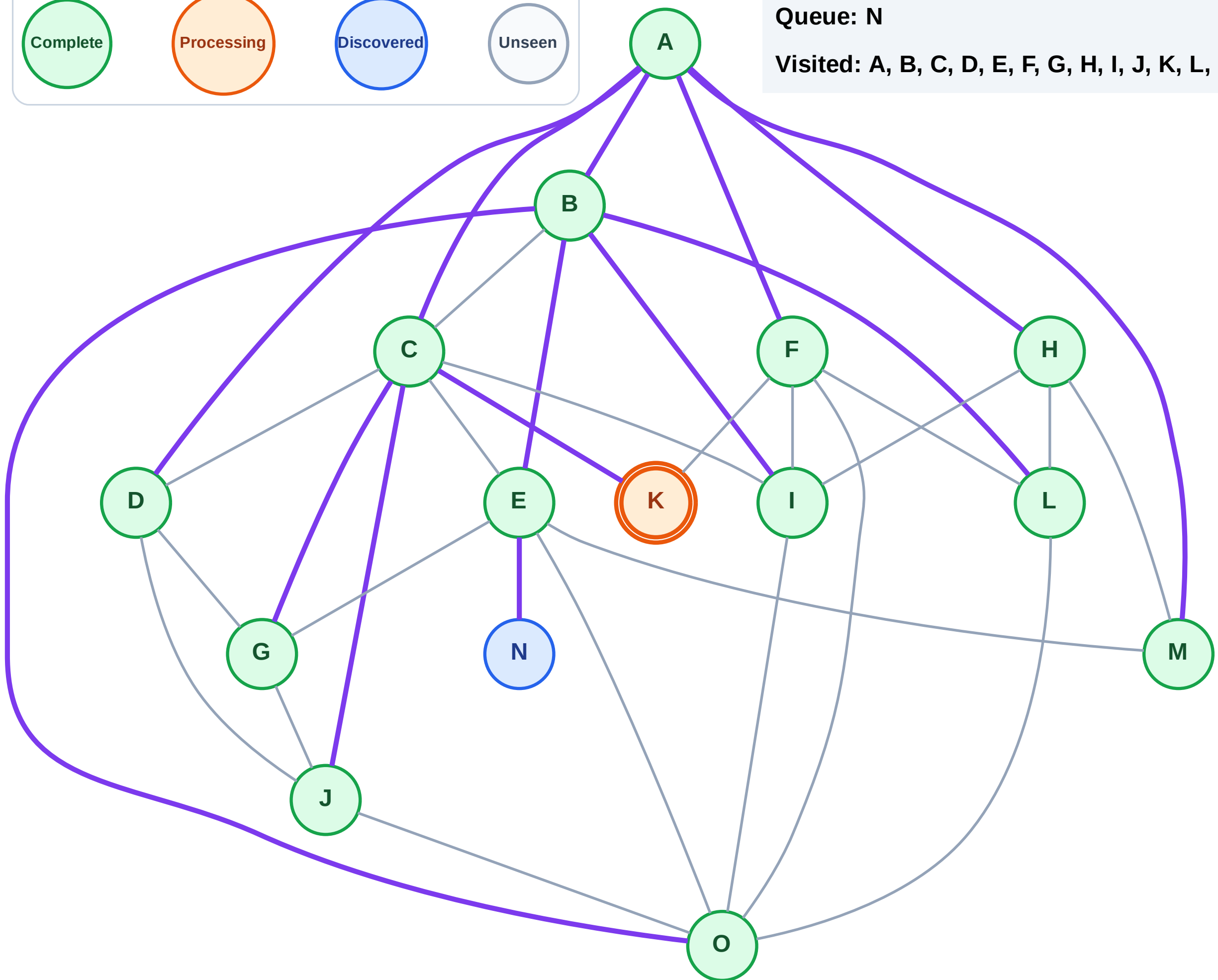
Step 29: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Queue: N

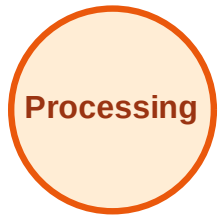
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



DFS Traversal

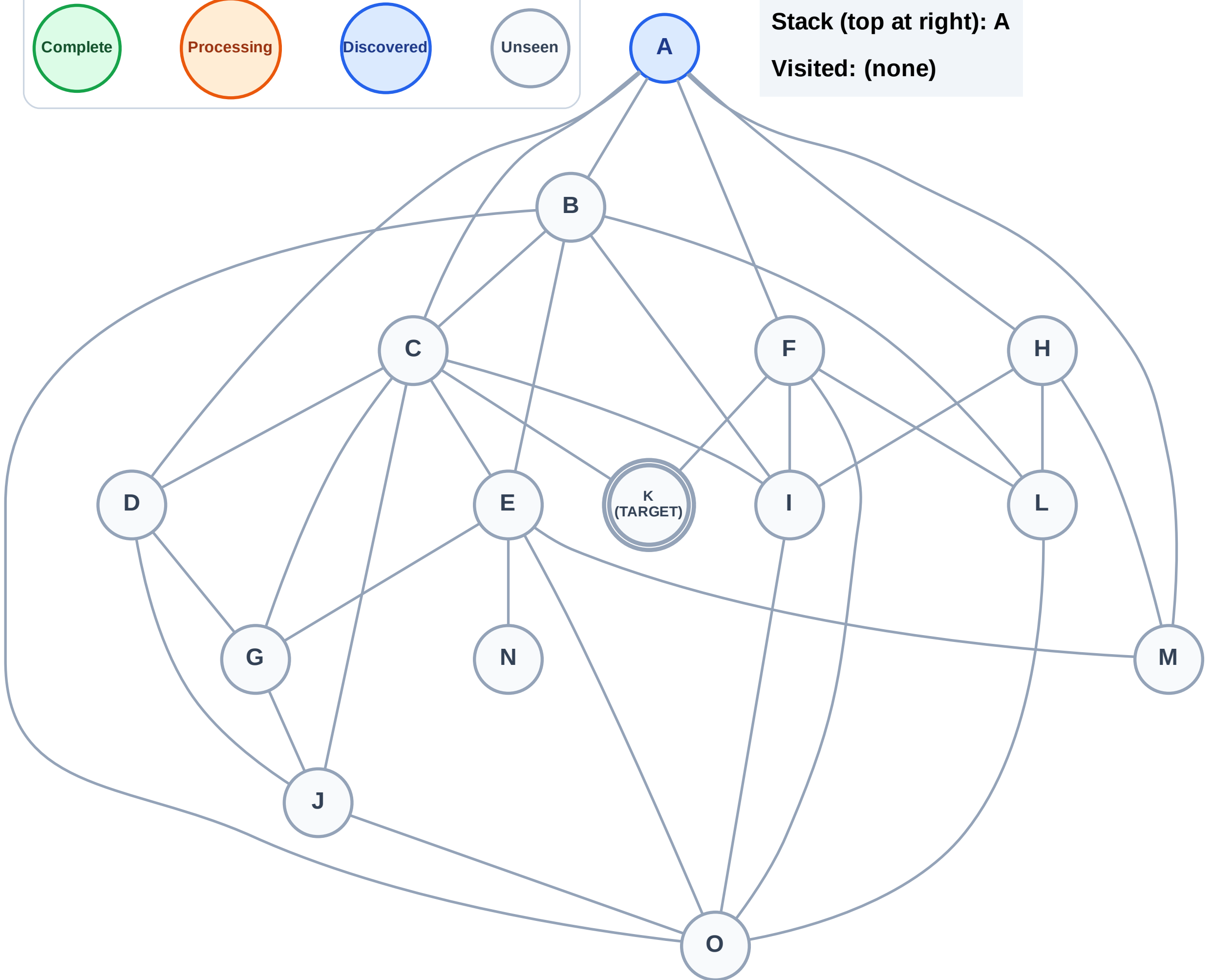
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

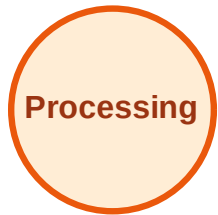
Visited: (none)



DFS Traversal

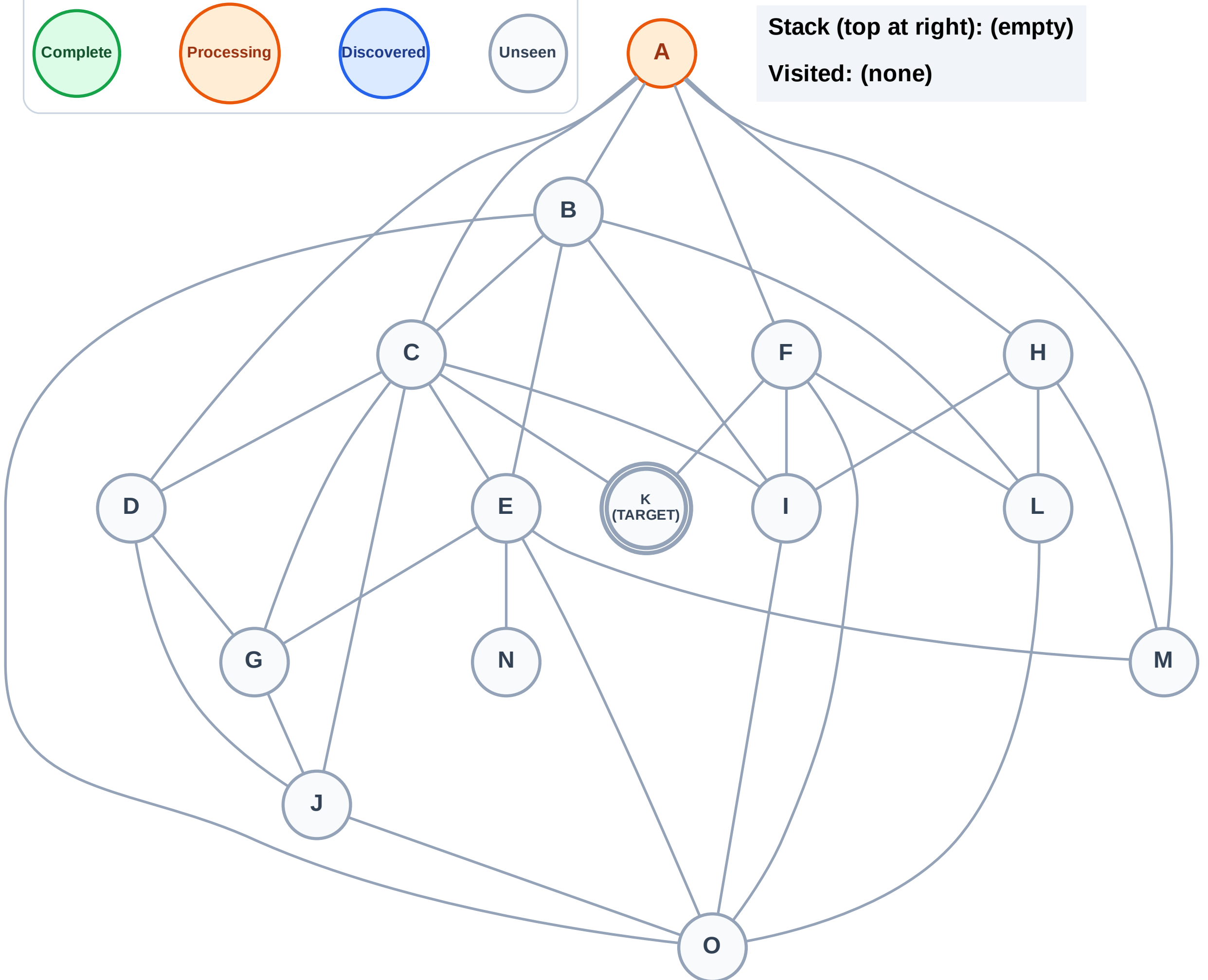
Step 2: Process node A

Legend



Stack (top at right): (empty)

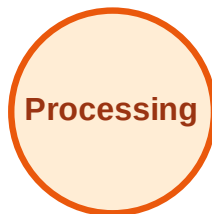
Visited: (none)



DFS Traversal

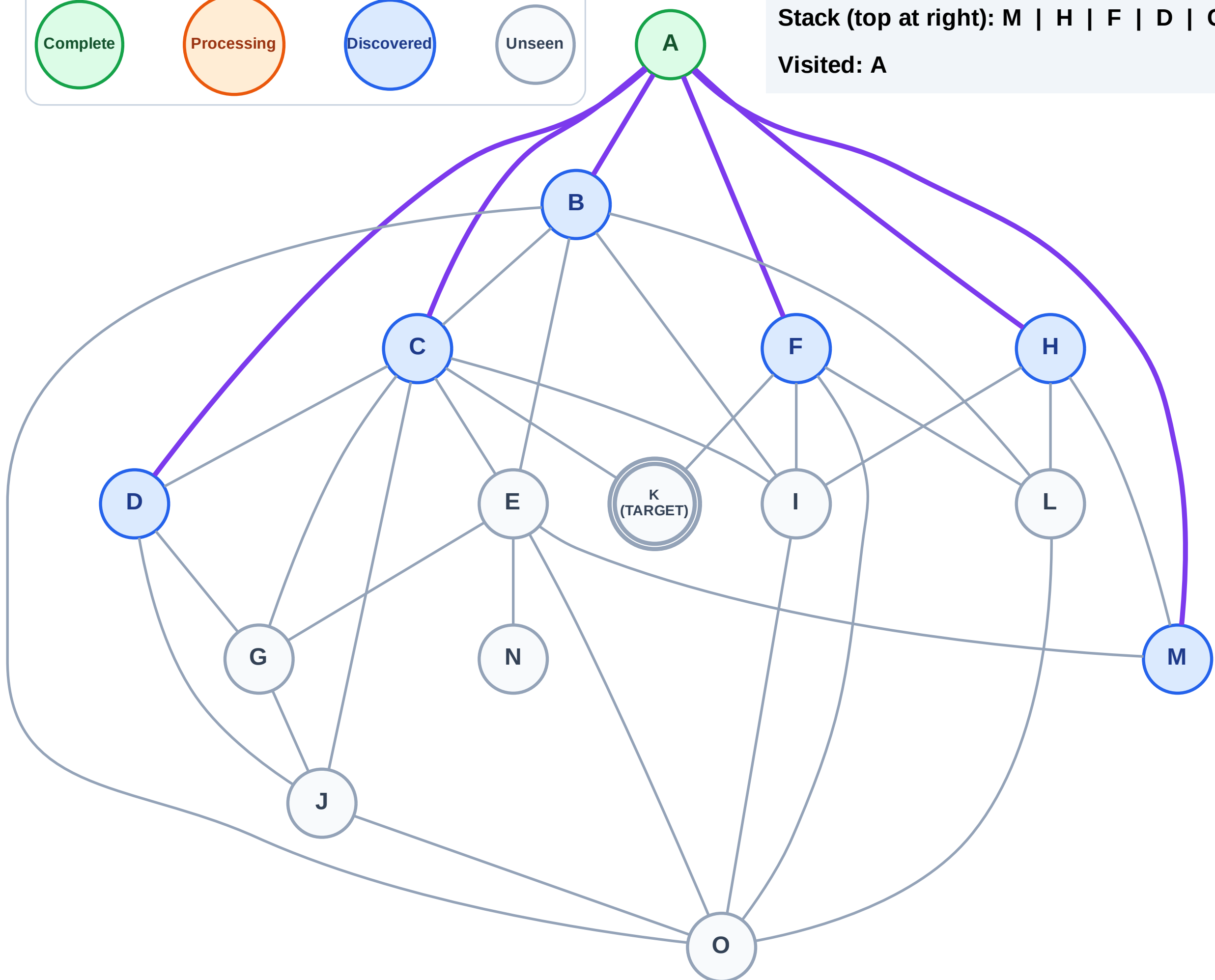
Step 3: Discover M, H, F, D, C, B from A

Legend



Stack (top at right): M | H | F | D | C | B

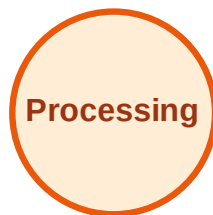
Visited: A



DFS Traversal

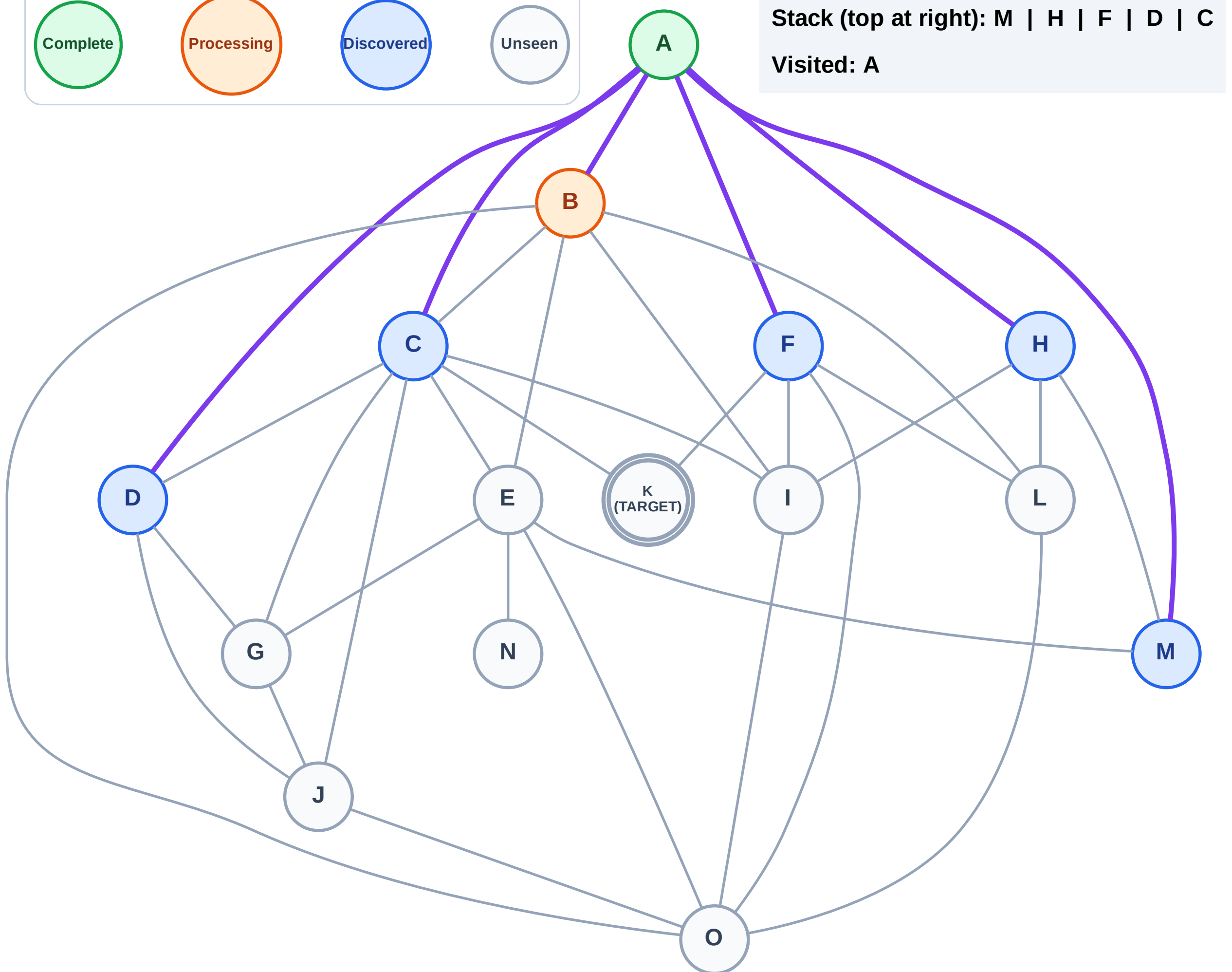
Step 4: Process node B

Legend



Stack (top at right): M | H | F | D | C

Visited: A



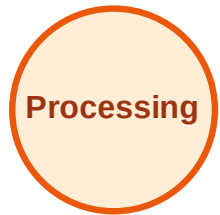
DFS Traversal

Step 5: Discover O, L, I, E from B

Legend



Complete



Processing



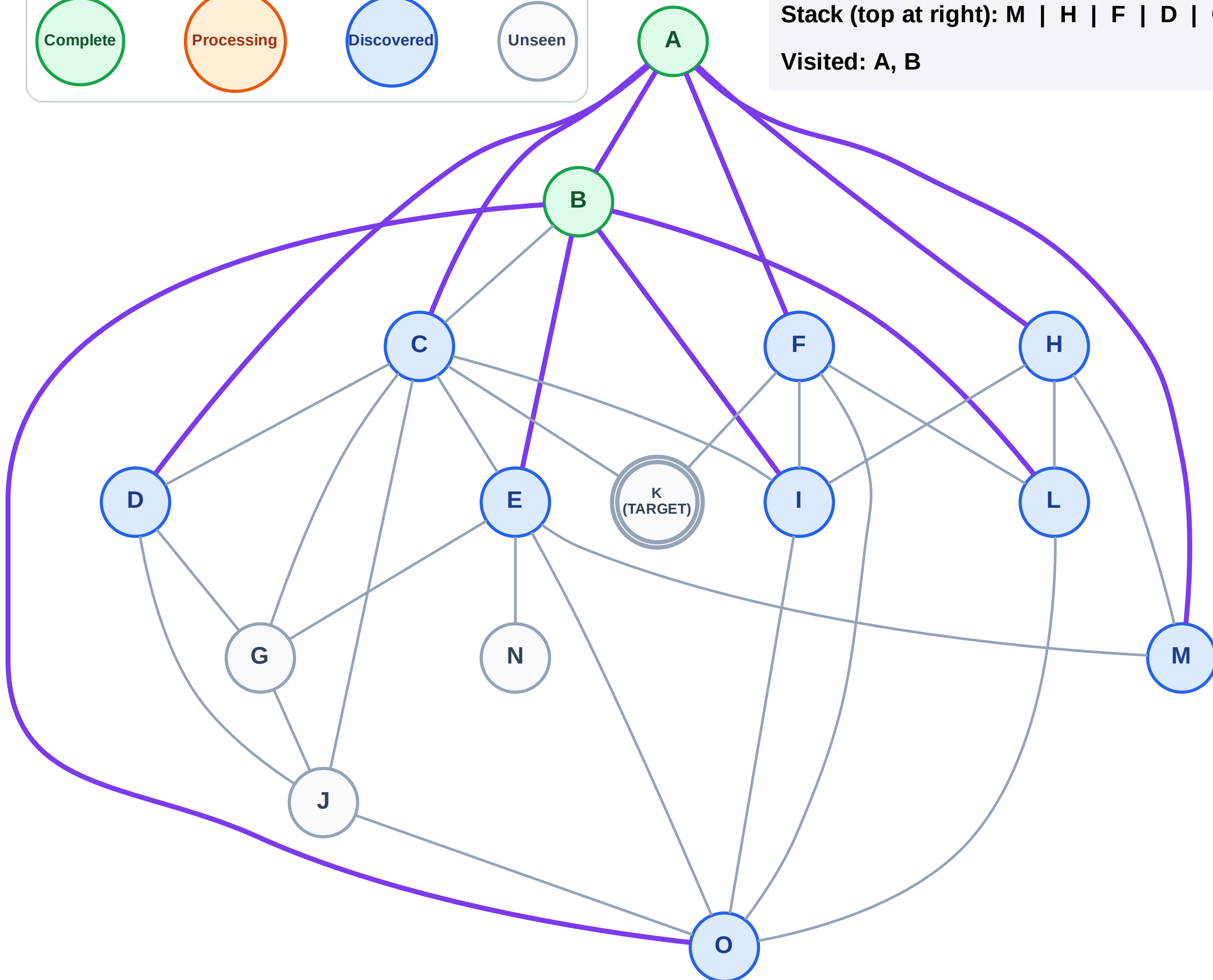
Discovered



Unseen

Stack (top at right): M | H | F | D | C | O | L | I | E

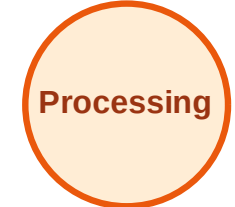
Visited: A, B



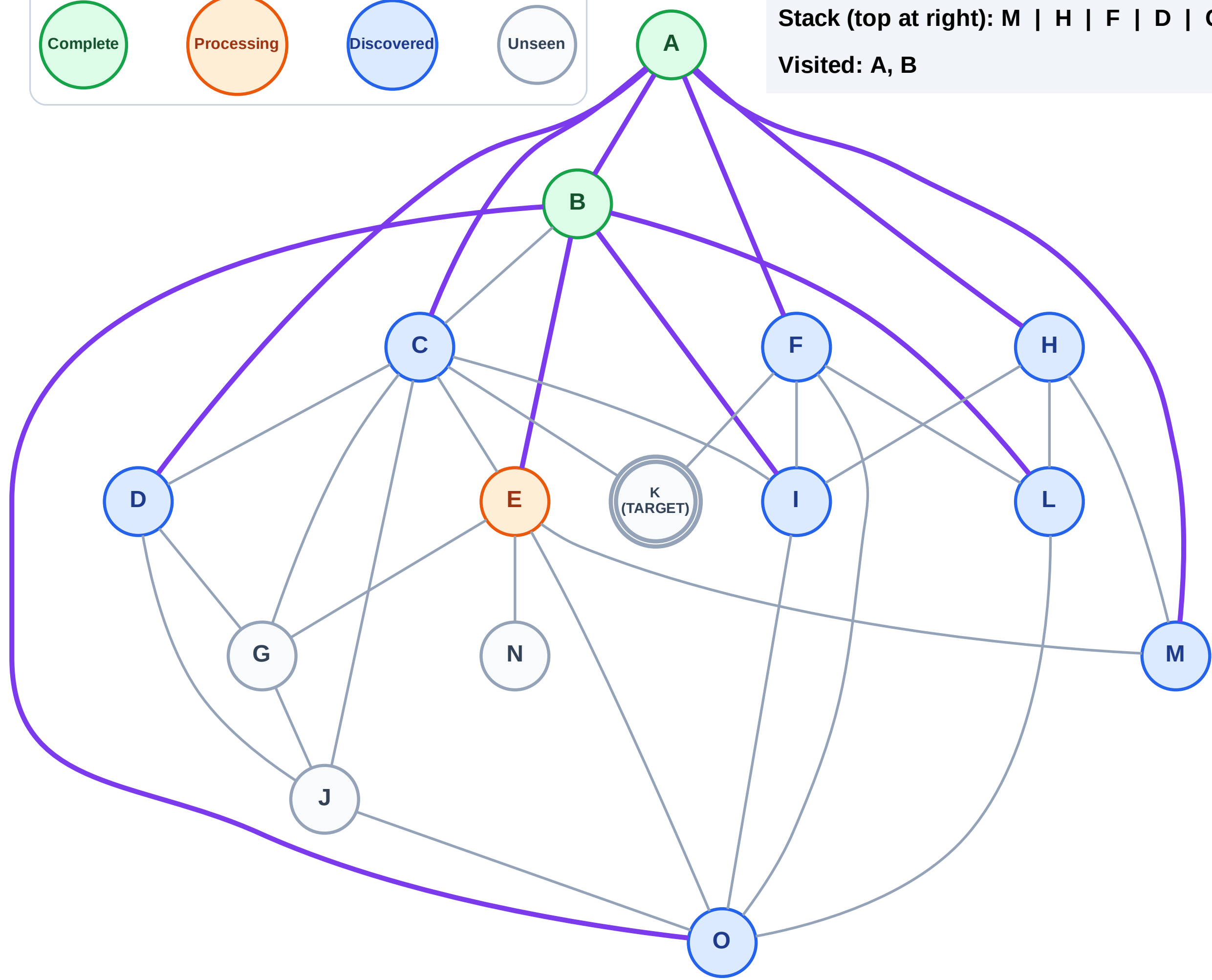
DFS Traversal

Step 6: Process node E

Legend



Stack (top at right): M | H | F | D | C | O | L | I
Visited: A, B



DFS Traversal

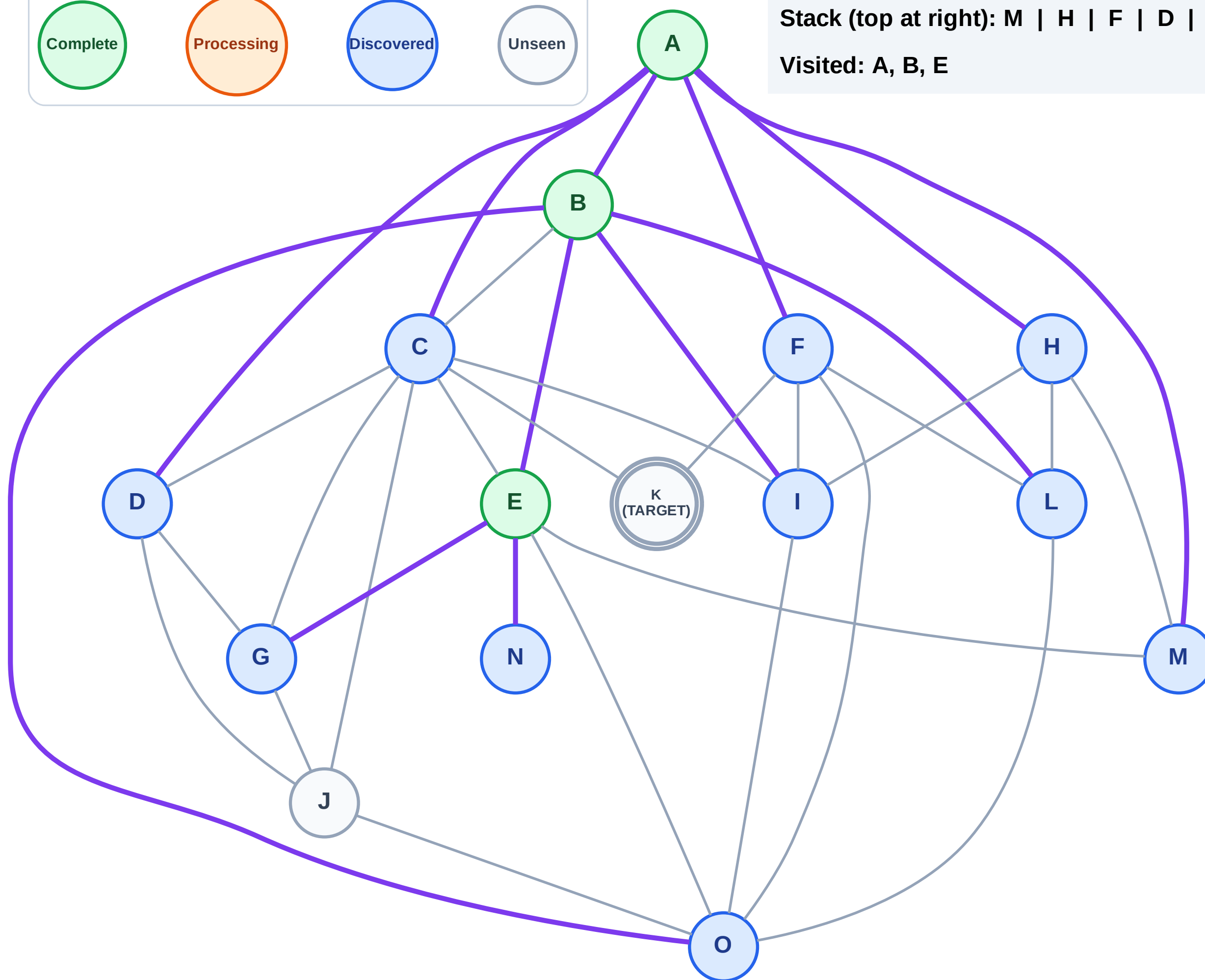
Step 7: Discover N, G from E

Legend



Stack (top at right): M | H | F | D | C | O | L | I | N | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend

Complete

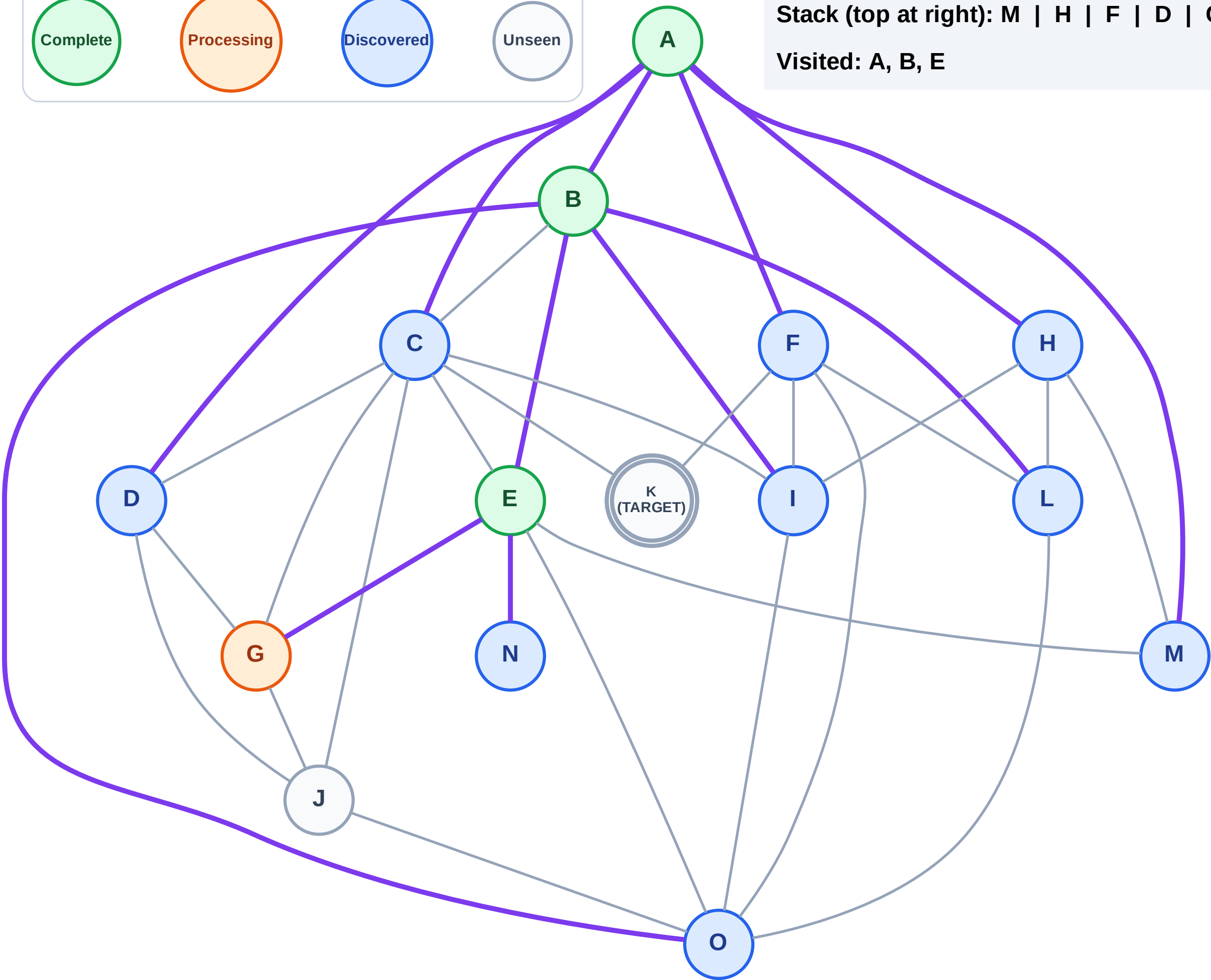
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L | I | N

Visited: A, B, E



DFS Traversal

Step 9: Discover J from G

Legend

Complete

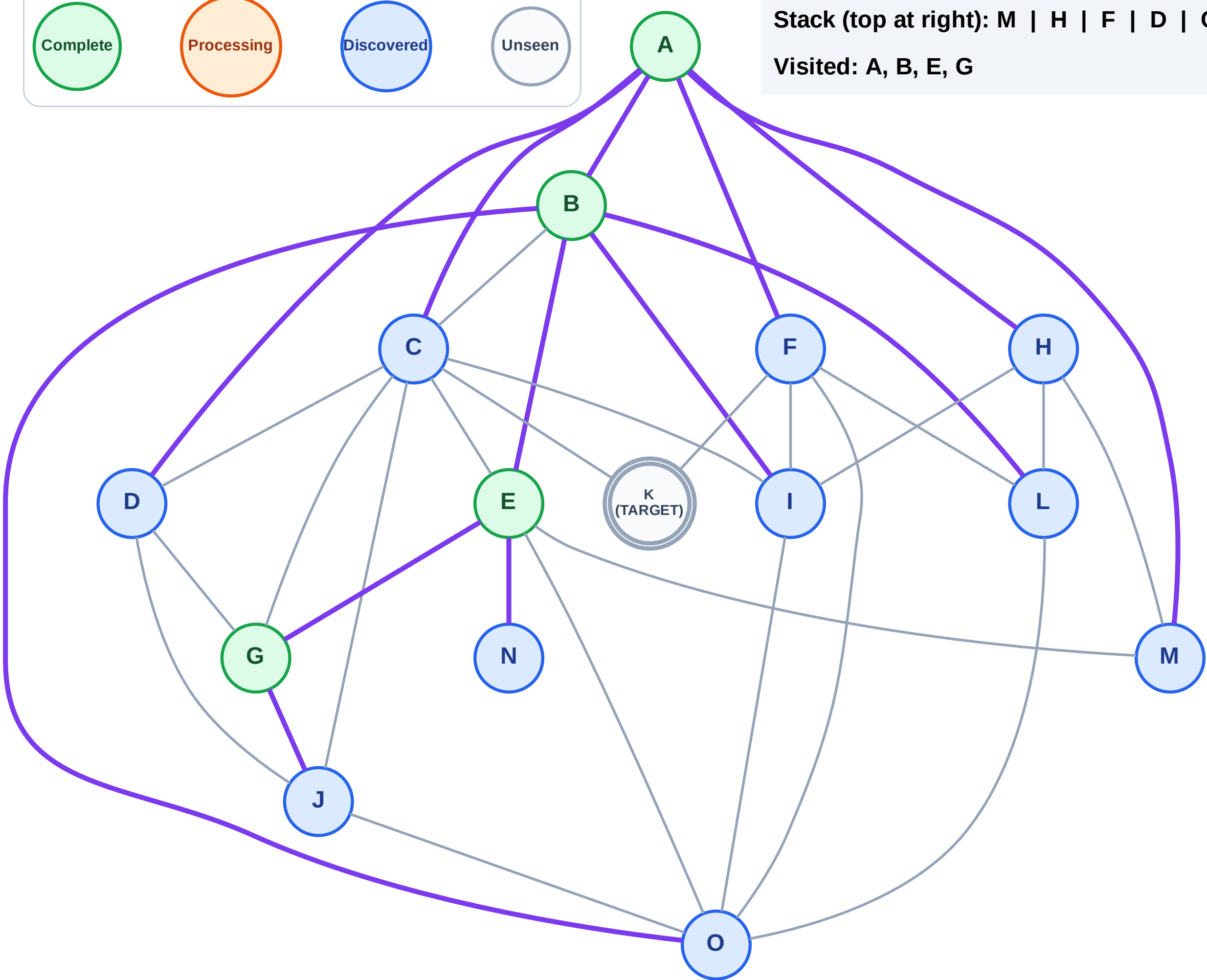
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L | I | N | J

Visited: A, B, E, G



DFS Traversal

Step 10: Process node J

Legend

Complete

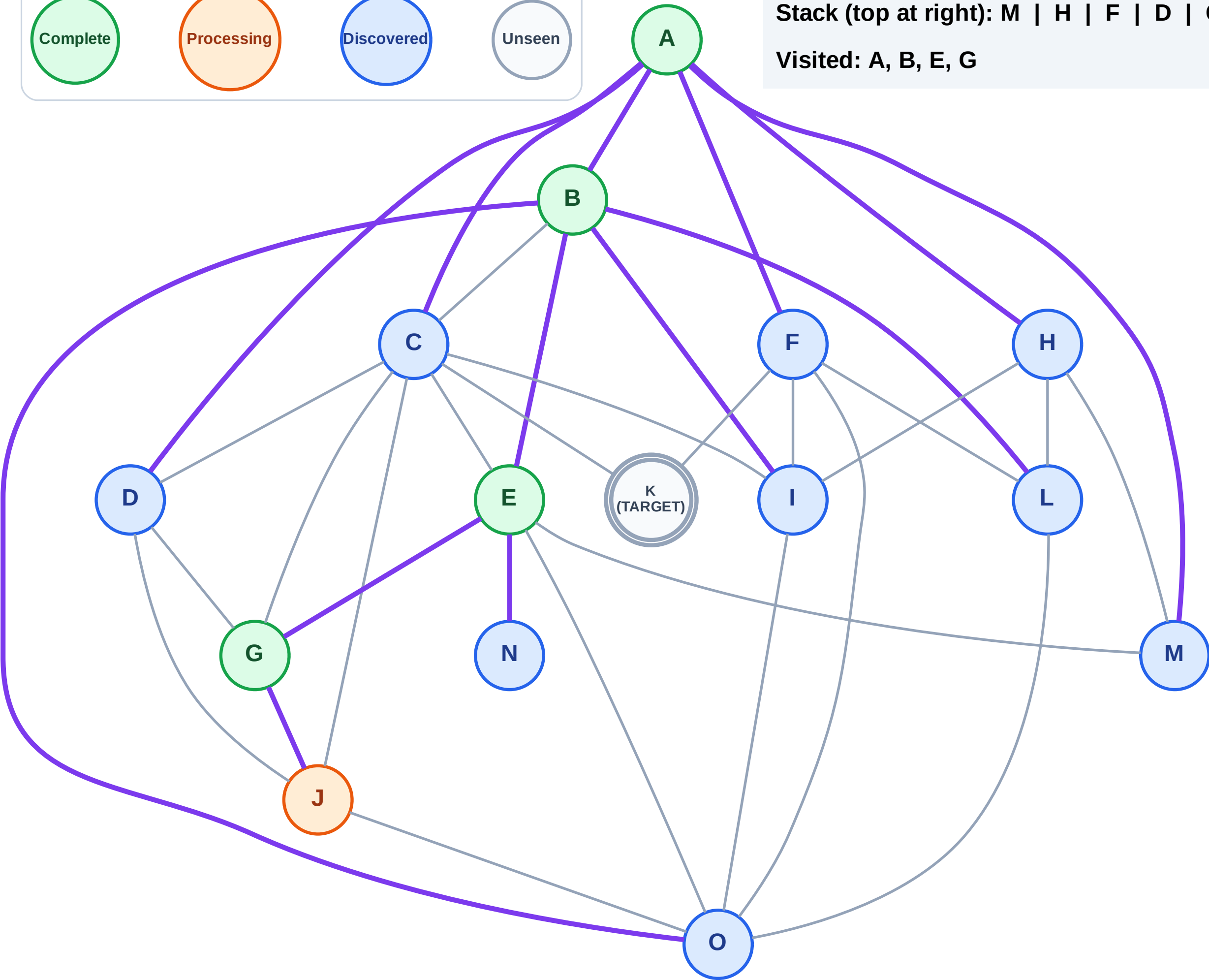
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L | I | N

Visited: A, B, E, G



DFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



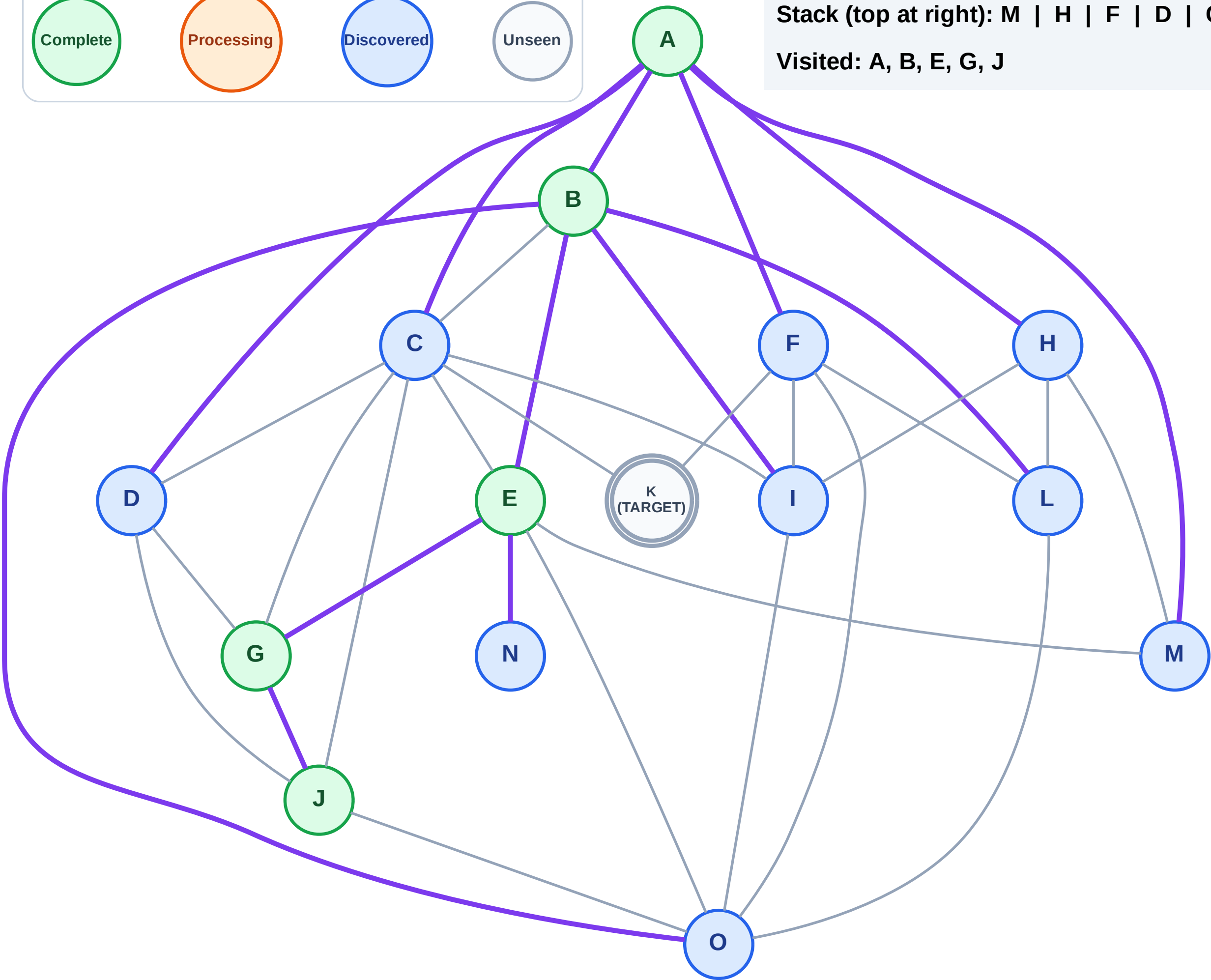
Discovered



Unseen

Stack (top at right): M | H | F | D | C | O | L | I | N

Visited: A, B, E, G, J



DFS Traversal

Step 12: Process node N

Legend

Complete

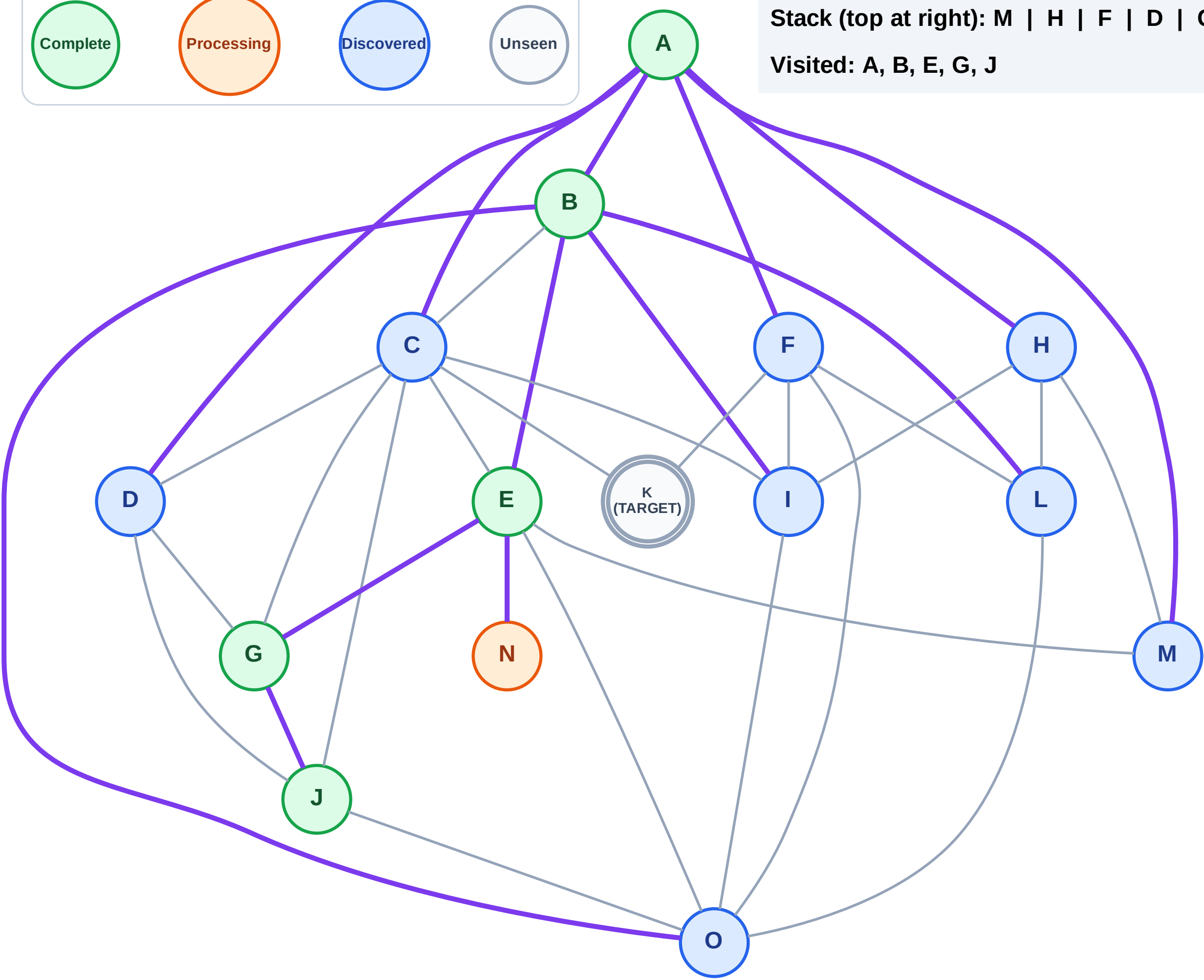
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L | I

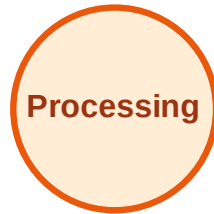
Visited: A, B, E, G, J



DFS Traversal

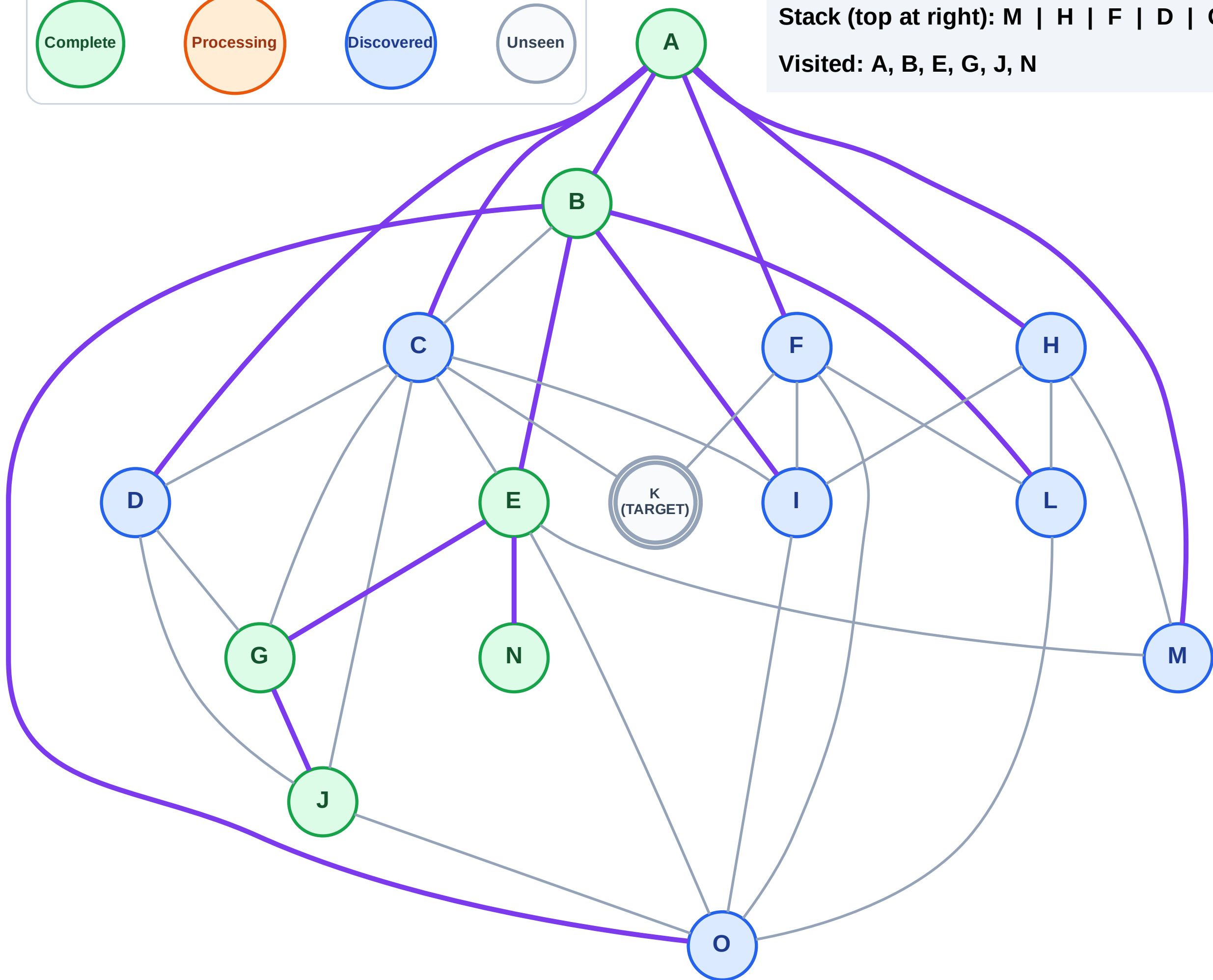
Step 13: No new neighbors from N

Legend



Stack (top at right): M | H | F | D | C | O | L | I

Visited: A, B, E, G, J, N



DFS Traversal

Step 14: Process node I

Legend

Complete

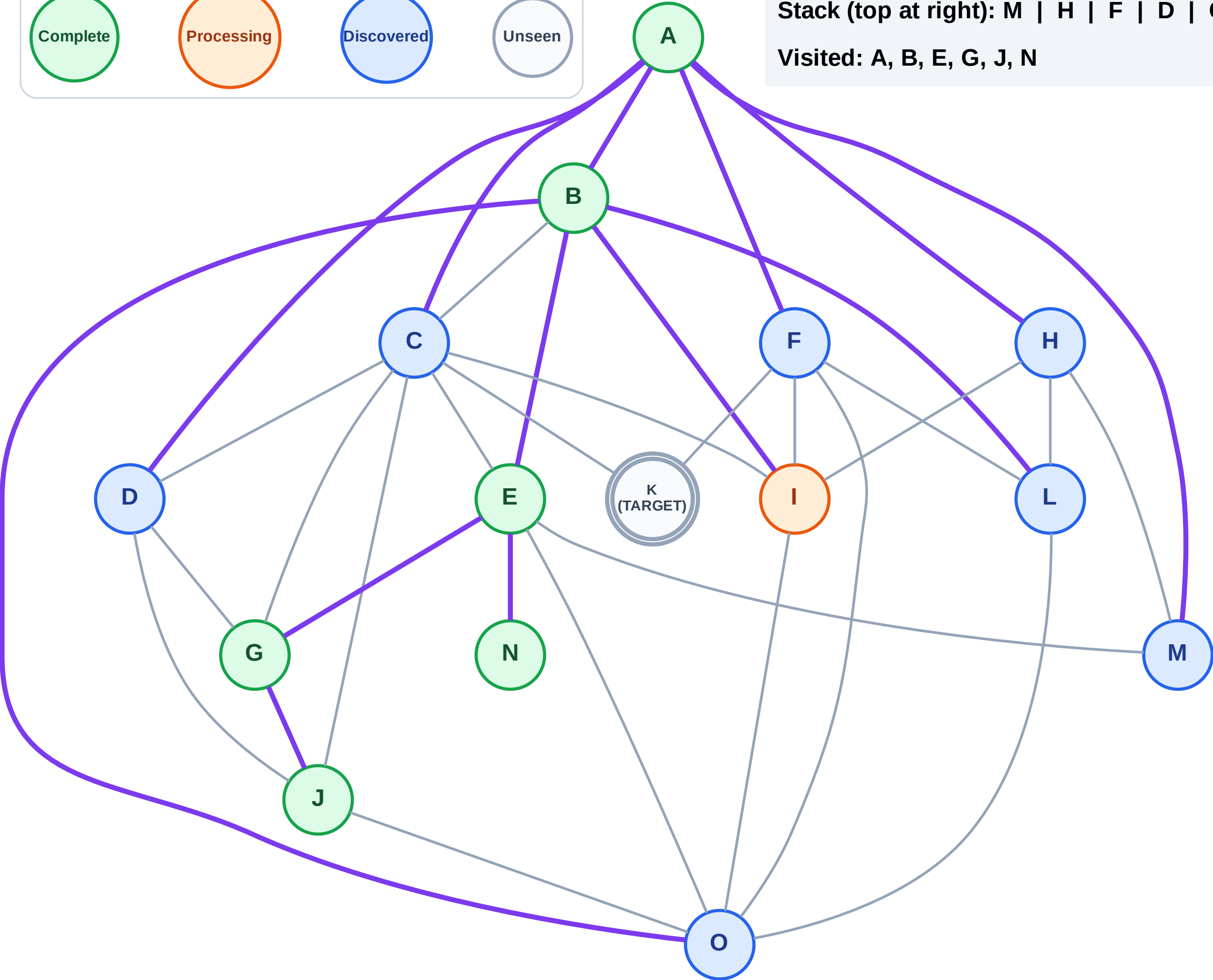
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L

Visited: A, B, E, G, J, N



DFS Traversal

Step 15: No new neighbors from I

Legend

Complete

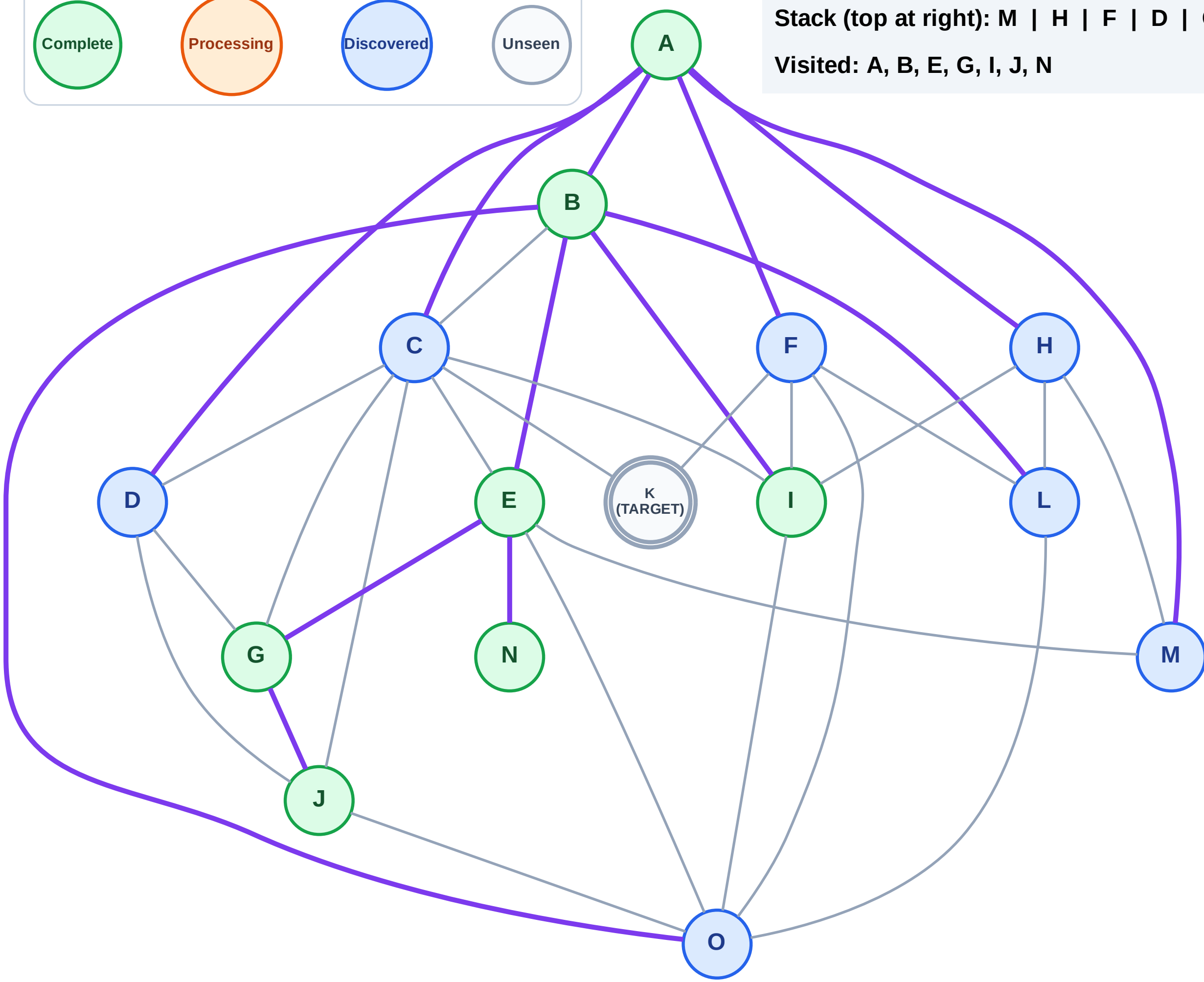
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D | C | O | L

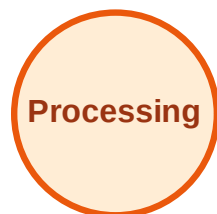
Visited: A, B, E, G, I, J, N



DFS Traversal

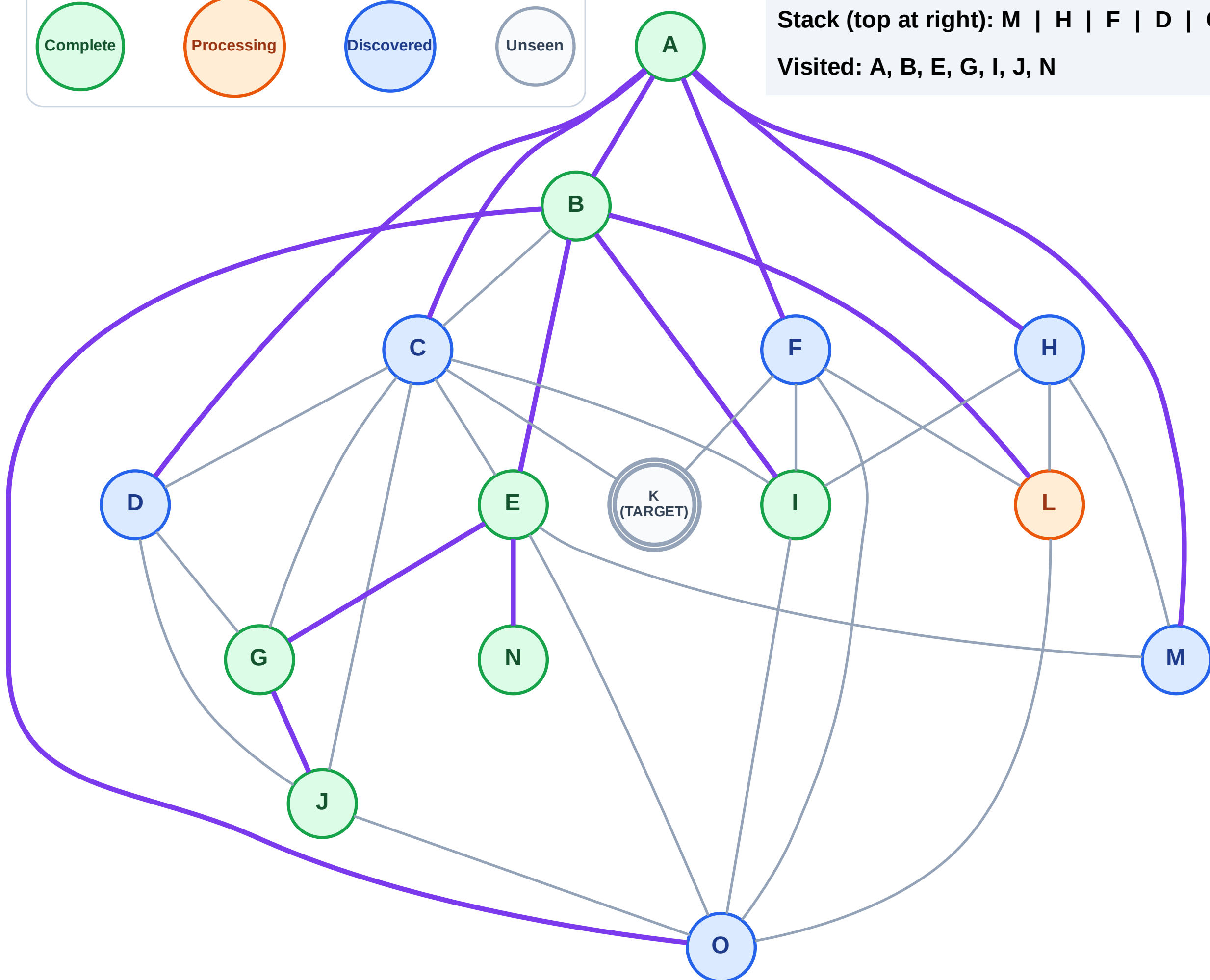
Step 16: Process node L

Legend



Stack (top at right): M | H | F | D | C | O

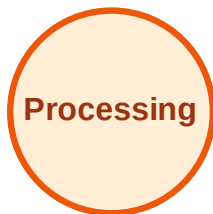
Visited: A, B, E, G, I, J, N



DFS Traversal

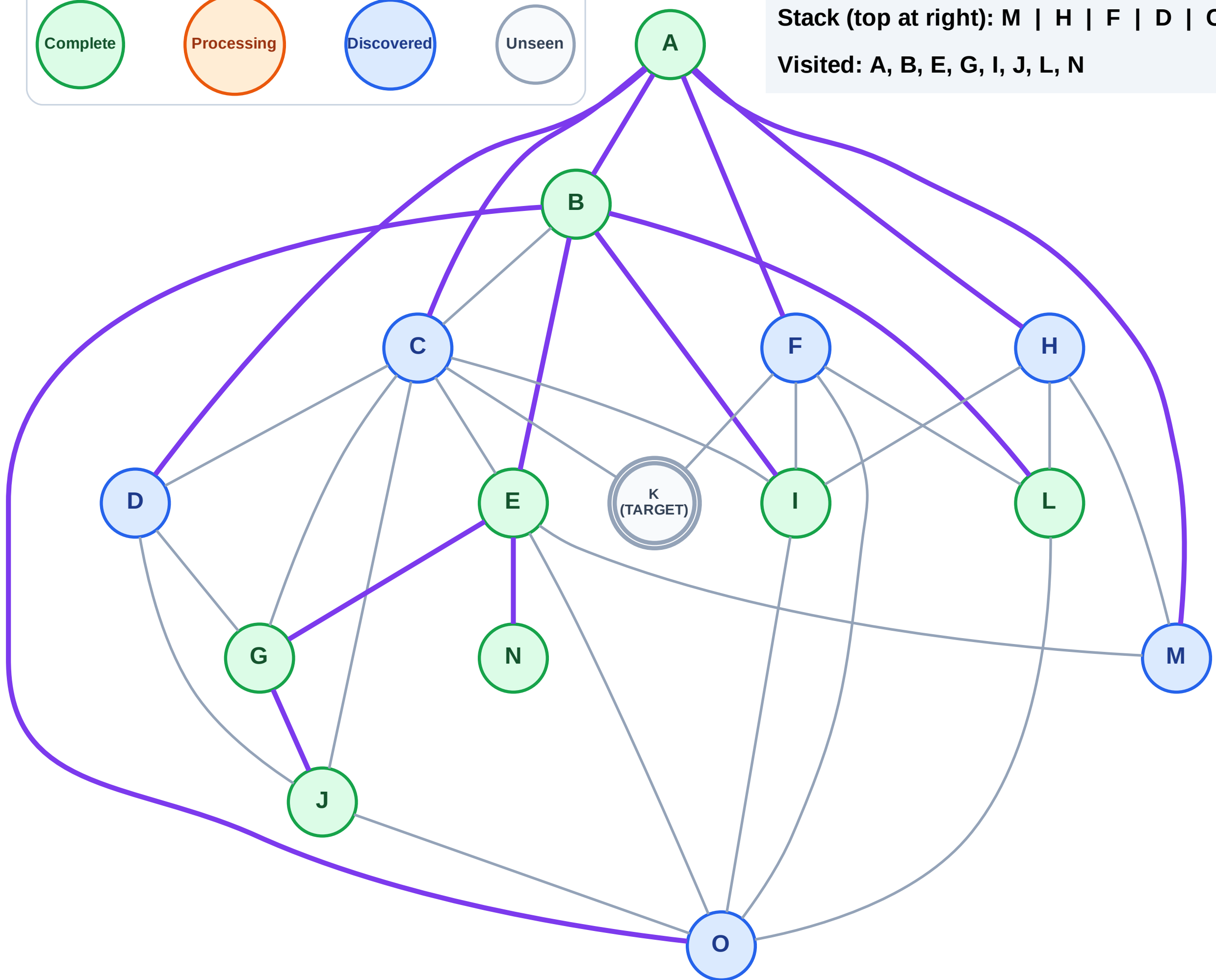
Step 17: No new neighbors from L

Legend



Stack (top at right): M | H | F | D | C | O

Visited: A, B, E, G, I, J, L, N



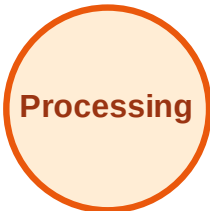
DFS Traversal

Step 18: Process node O

Legend



Complete



Processing



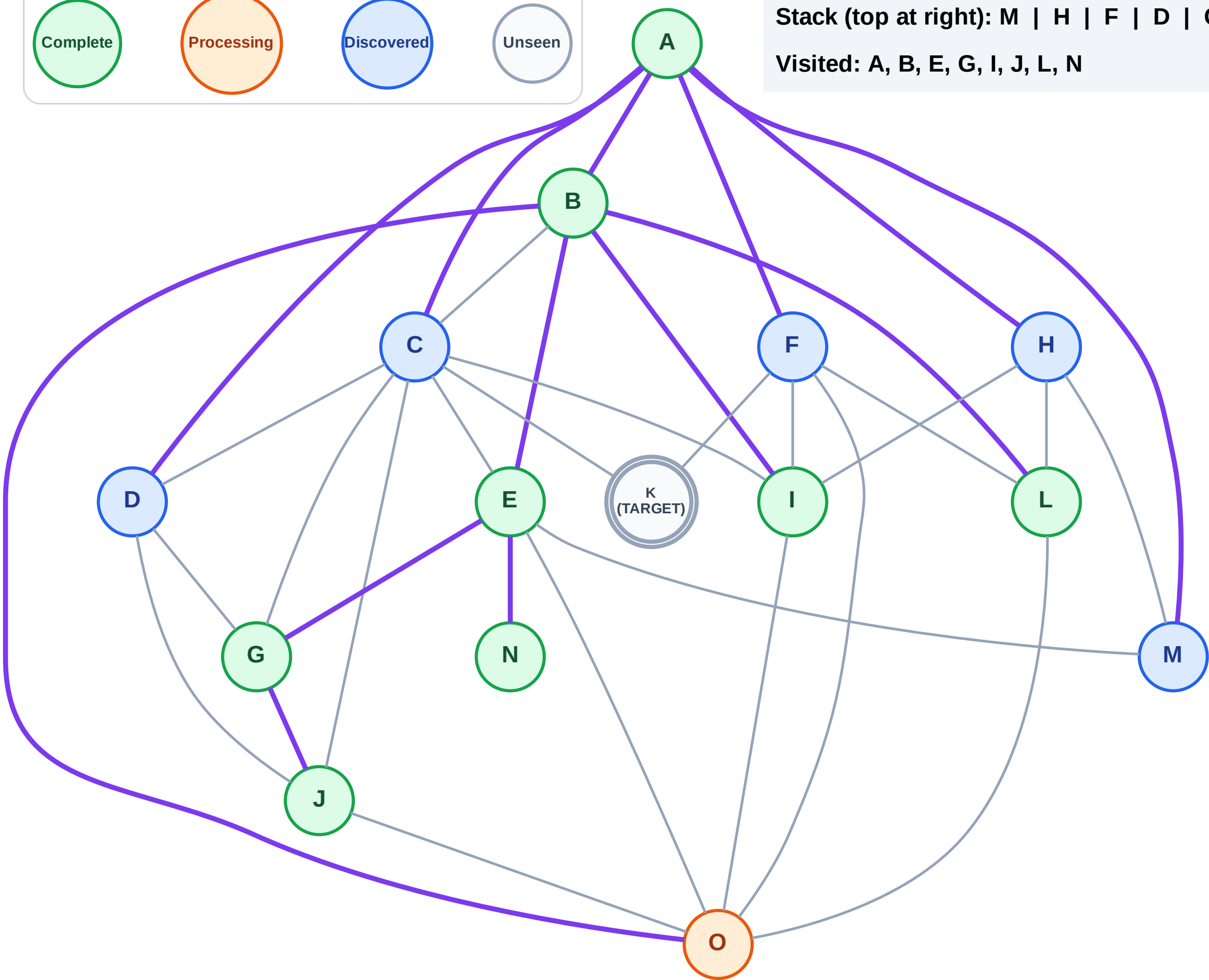
Discovered



Unseen

Stack (top at right): M | H | F | D | C

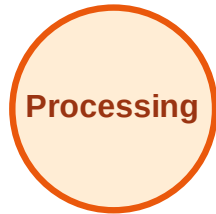
Visited: A, B, E, G, I, J, L, N



DFS Traversal

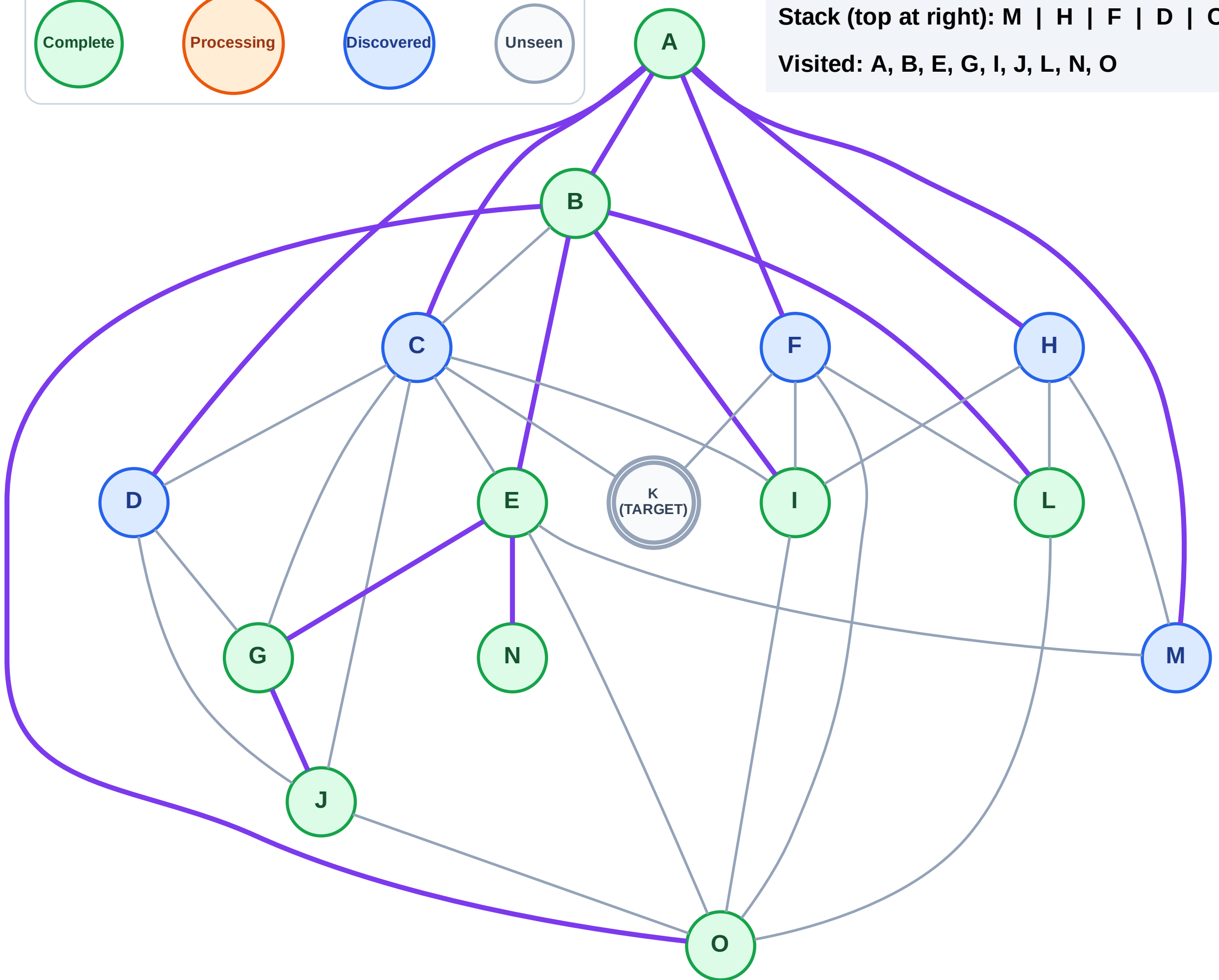
Step 19: No new neighbors from O

Legend



Stack (top at right): M | H | F | D | C

Visited: A, B, E, G, I, J, L, N, O



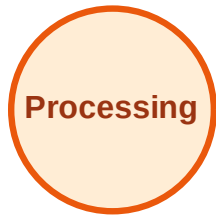
DFS Traversal

Step 20: Process node C

Legend



Complete



Processing



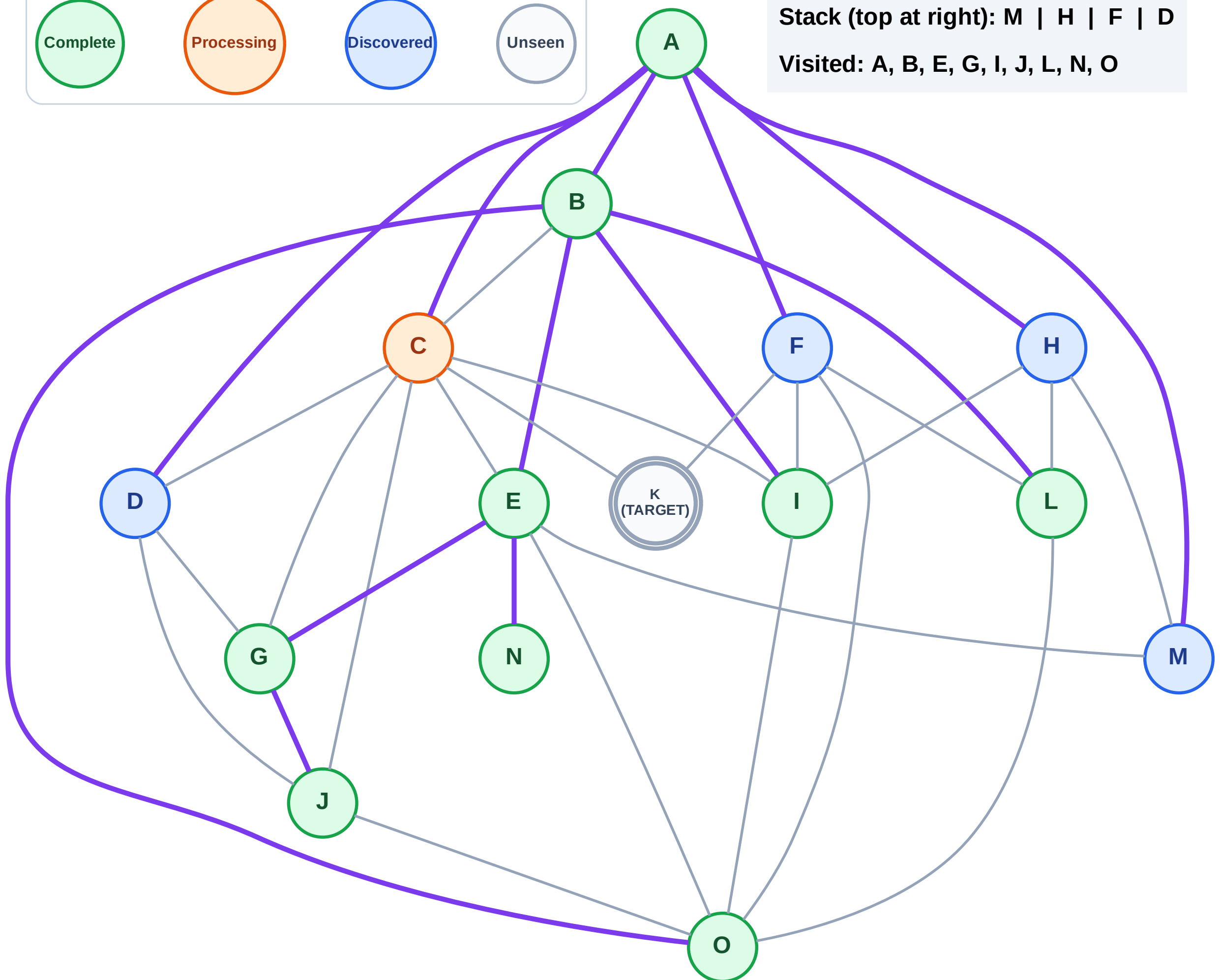
Discovered



Unseen

Stack (top at right): M | H | F | D

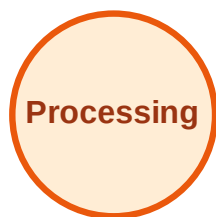
Visited: A, B, E, G, I, J, L, N, O



DFS Traversal

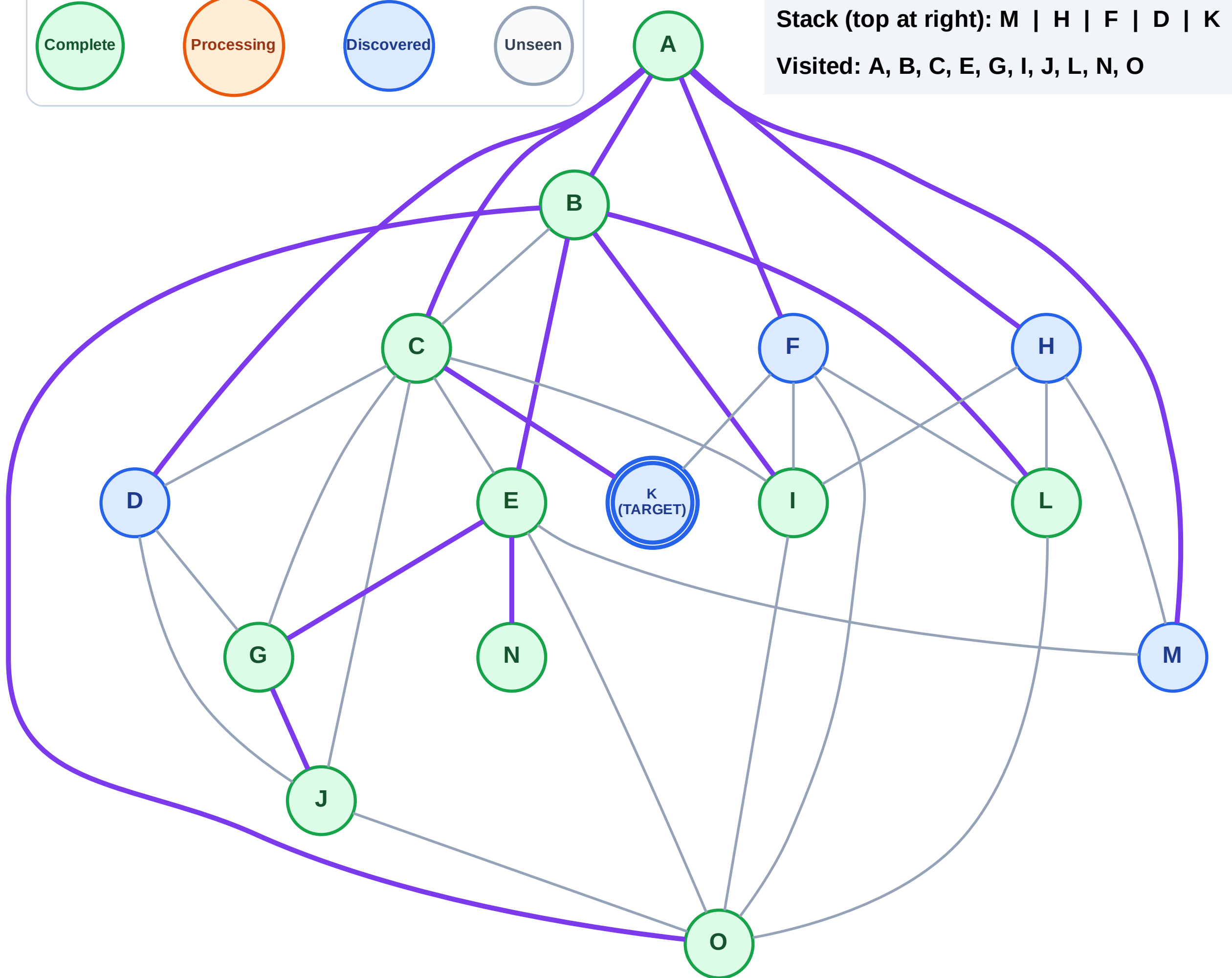
Step 21: Discover K from C

Legend



Stack (top at right): M | H | F | D | K

Visited: A, B, C, E, G, I, J, L, N, O



DFS Traversal

Step 22: Process node K

Legend

Complete

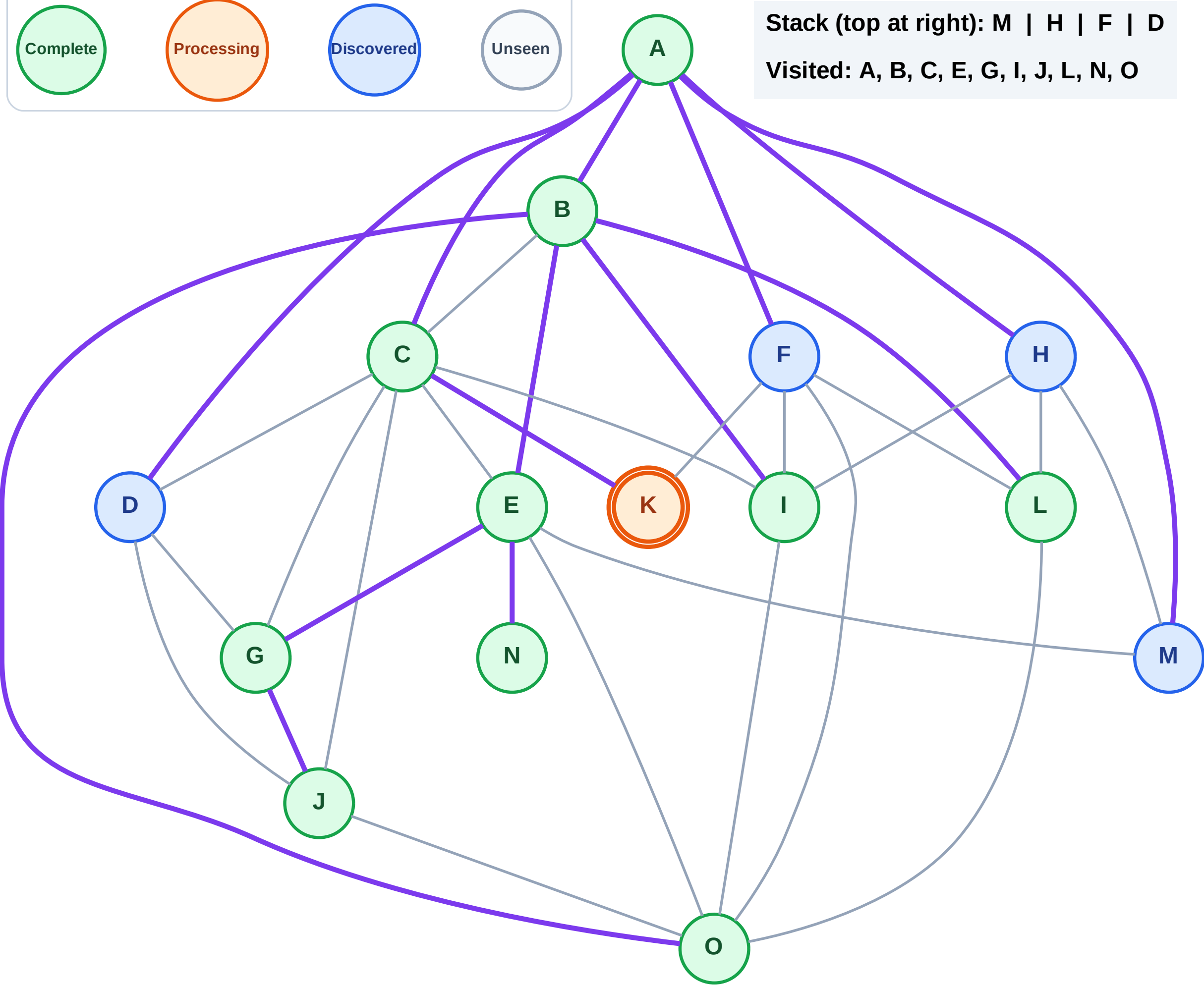
Processing

Discovered

Unseen

Stack (top at right): M | H | F | D

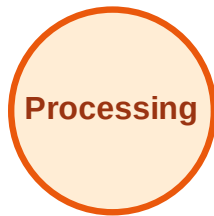
Visited: A, B, C, E, G, I, J, L, N, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Stack (top at right): M | H | F | D

Visited: A, B, C, E, G, I, J, K, L, N, O

